

In-house Project Report (End Term Examination)

titled

In Silico Comparative Studies of Bacoside A and Bacoside B

completed at

**MITADTU School of Bioengineering Sciences & Research,
MIT ADT University**

by

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Under the guidance of

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**IN PARTIAL FULFILMENT FOR THE DEGREE OF
INTEGRATED MASTER OF TECHNOLOGY IN BIOENGINEERING**

Submitted to

School of Bioengineering Sciences & Research,
A Constituent Unit of MIT-ADT University, Pune.

2026 - 2027

CERTIFICATE

This is to certify that **Ms. Pranjal Salil Shekhar (MITU22IMBI0017)** has successfully completed the in-house project entitled "**In Silico Comparative Studies of Bacoside A and Bacoside B**" [SP/2025-2026/003] under the guidance of Dr. Sanket Bapat at School of Bioengineering Sciences & Research, a constituent unit of MIT-ADT University, Pune, during the academic year 2026-2027, for the partial fulfillment for the award of the degree of Integrated M.Tech in Bioengineering.

Date: 29/4/2026

Place: Pune

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Dr. Sanket Bapat
Mentor / Guide

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Head of the School

DECLARATION

I declare that this report titled "**In Silico Comparative Studies of Bacoside A and Bacoside B**" [SP/2025-2026/003] reflects my original work about the subject in my own words. I have sufficiently cited and referenced the original sources, referred or considered in this work. I have not plagiarized or submitted the same work for the award of any other degree. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute.

.....
Pranjal Salil Shekhar
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Date: 29/04/2026

Place: Pune

School of Bioengineering Sciences & Research,
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ABSTRACT

Bacopa monnieri (Brahmi), a well-established medicinal plant of the Plantaginaceae family, is renowned in Ayurvedic medicine for its nootropic, neuroprotective, antioxidant, and adaptogenic properties. Its two primary bioactive constituents: Bacoside A and Bacoside B are triterpenoid saponins that have been the subject of extensive scientific investigation, although research has been highly asymmetric, focusing predominantly on Bacoside A while leaving Bacoside B comparatively underexplored.

The present study undertakes a comprehensive comparative in silico molecular docking analysis of Bacoside A and Bacoside B against cancer hub proteins associated with Glioblastoma Multiforme (GBM) and Hepatocellular Carcinoma (liver cancer). Protein targets were selected based on network pharmacology analysis and retrieved from the RCSB Protein Data Bank (PDB). Ligand structures were obtained from PubChem, converted to 3D form using the Corina tool, and optimized for docking. Molecular docking was performed using AutoDock Vina, and protein-ligand complexes were visualized using PyMOL and BIOVIA Discovery Studio. Probable ligand binding sites were predicted using the P2Rank machine learning tool.

Results reveal that GBM cancer targets exhibit stronger binding affinities for Bacoside A, with the highest score of -9.9 kcal/mol observed at the 1IVO crystal structure (EGFR tyrosine kinase domain), and a P2Rank pocket score of 7.60. Hepatocellular carcinoma targets demonstrate stronger binding affinities for Bacoside B, with the highest score of -11.2 kcal/mol observed at 3O96 (AKT1/Protein Kinase B), and a P2Rank pocket score of 28.75. This study provides the first systematic computational evidence of Bacoside B's multi-target anticancer potential, and constitutes a scientific foundation for future in vitro and in vivo validation studies.

Keywords: *Bacopa monnieri*, Bacoside A, Bacoside B, Molecular Docking, Glioblastoma, Hepatocellular Carcinoma, AutoDock Vina, In Silico, Cancer Hub Proteins

CONTENTS

<i>Certificate</i>	<i>i</i>
<i>Declaration</i>	<i>ii</i>
<i>Abstract</i>	<i>iii</i>
<i>Contents</i>	<i>iv</i>
<i>List of Figures</i>	<i>v</i>
<i>List of Tables</i>	<i>vi</i>
<i>List of Abbreviations</i>	<i>vii</i>
Section 1: Introduction and Literature Review	9-14
1.1 <i>Bacopa monnieri: Ethnobotanical and Clinical Background</i>	
1.2 <i>Bacosides: The Primary Bioactive Constituents</i>	
1.3 <i>Need for Comparative Study and Research Gap</i>	
1.3 <i>Why Bacoside B Has Not Been Studied Until Now</i>	
1.5 <i>Cancer as a Multi-Target Disease: Rationale for This Study</i>	
Section 2: Methodology	15-17
2.1 <i>Literature Review and Target Selection</i>	
2.2 <i>Ligand Preparation</i>	
2.3 <i>Target Protein Selection and Preparation</i>	
2.4 <i>Binding Site Prediction (P2Rank)</i>	
2.5 <i>Molecular Docking Protocol</i>	
2.6 <i>Visualization and Interaction Analysis</i>	
2.7 <i>Molecular Dynamics Simulation and ADMET Analysis</i>	
Section 3: Results and Discussion	18-67
3.1 <i>Docking Analysis Overview</i>	
3.2 <i>Glioblastoma Multiforme (GBM)</i>	
3.3 <i>Protein Ligand interaction analysis using Discovery Studio</i>	
3.4 <i>HEPATOCELLULAR CARCINOMA (HCC)</i>	
3.5 <i>Protein Ligand interaction analysis using Discovery Studio</i>	
3.6 <i>Discussion: Pharmacological Implications</i>	
Section 4: Conclusion	68-69
Section 5: References and Bibliography	70

LIST OF FIGURES

Fig. No.	Details	Page No.
1.1	<i>Bacopa Monnieri Plants</i>	9
1.2.1	<i>Phytochemicals of Bacopa Monnieri</i>	10
1.2.2	<i>Major saponins of Bacopa Monnieri</i>	10
1.2.1.1	Structure of Bacoside A	11
1.2.1.2	Composition of Bacoside A	11
1.2.2.1	Structure of Bacoside B	12
2.1	<i>Methodology work flow</i>	15
3.2.1	Probable Pocket Binding Sites of 1IVO (P2Rank)	19
4	Probable Pocket Binding Sites of 3O96 (P2Rank)	47
5	Binding Affinity and Docking Sites — GBM Proteins (1IVO, 4X17, 1YWN, 5A63, 1GNG, 2YZQ, 1YC5, 3IQJ, 4XBM, 1CTQ, 2MH3, 2DB7, 2Z6H, 6BD4)	19-28
6	Binding Affinity and Docking Sites — Liver Cancer Proteins (1TUP, 1P9M, 3O96, 2H5I, 2AZ5, 2BDJ, 3B28, 3C7Q, 1M17)	48-53
7	BIOVIA Discovery Studio 2D Interaction Maps — All 42 Complexes	29-65

LIST OF TABLES

Table No.	Details	Page No.
1	Cancer Hub Proteins: PDB IDs, Biological Roles, and Cancer Relevance	16
2	Binding Free Energies of Bacoside A and B(GBM)	18
3	Interaction Comparison at 1IVO	29
4	Interaction Comparison at 4X17	30
5	Interaction Comparison at 1YWN	31
6	Interaction Comparison at 5A63	33
7	Interaction Comparison at 1GNG	34
8	Interaction Comparison at 2YZQ	35
9	Interaction Comparison at 1YC5	36
10	Interaction Comparison at 3IQJ	38
11	Interaction Comparison at 4XBM	39
12	Interaction Comparison at 1CTQ	40
13	Interaction Comparison at 2MH3	41
14	Interaction Comparison at 2DB7	43
15	Interaction Comparison at 64BD	44
16	Interaction Comparison at 2ZH6	45
17	Comparative Docking Binding Free Energies of Bacoside A and B(HCC)	47
18	Interaction Comparison at AKT1 (3O96)	54
19	Interaction Comparison at TP53 (1TUP)	55
20	Interaction Comparison at EGFR (1M17)	57
21	Interaction Comparison at TNF (2AZ5)	58
22	Interaction Comparison at IL6 (1P9M)	60
23	Interaction Comparison at VEGFA (3C7Q)	61
24	Interaction Comparison at HSP90AA1(3B28)	62
25	Interaction Comparison at SRC (2BDJ)	63
26	Interaction Comparison at CASP3 (2H5I)	65

LIST OF ABBREVIATIONS

Abbreviation	Full Form
GBM	Glioblastoma Multiforme
HCC	Hepatocellular Carcinoma
PDB	Protein Data Bank
EGFR	Epidermal Growth Factor Receptor
AKT1	RAC-alpha Serine/Threonine-protein Kinase / Protein Kinase B
TP53	Tumor Protein p53
TNF	Tumor Necrosis Factor
IL6	Interleukin-6
VEGFA	Vascular Endothelial Growth Factor A
HSP90AA1	Heat Shock Protein 90 Alpha Family Class A Member 1
SRC	Proto-oncogene Tyrosine-protein Kinase Src
CASP3	Caspase-3
MD	Molecular Dynamics
ADMET	Absorption, Distribution, Metabolism, Excretion, Toxicity
SMILES	Simplified Molecular Input Line Entry System
NMR	Nuclear Magnetic Resonance
HPLC	High Performance Liquid Chromatography
TLC	Thin Layer Chromatography
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (cell viability assay)
SPR	Surface Plasmon Resonance
ITC	Isothermal Titration Calorimetry
EMT	Epithelial-Mesenchymal Transition
PI3K	Phosphoinositide 3-Kinase
IAP	Inhibitor of Apoptosis Protein

SECTION 1: INTRODUCTION AND LITERATURE REVIEW

1.1 *Bacopa monnieri*: Ethnobotanical and Clinical Background

Bacopa monnieri, commonly known as Brahmi or Water Hyssop, is a small creeping perennial herb widely distributed in marshy and wetland areas of India, Nepal, Sri Lanka, China, and other tropical regions. It belongs to the family Plantaginaceae and has been used for more than 3,000 years in the Ayurvedic system of medicine under the classification of "Medhya Rasayana" a group of herbs specifically intended to rejuvenate the intellect and consciousness. Ancient Ayurvedic texts including the Charaka Samhita and Sushruta Samhita reference Brahmi as a potent brain tonic capable of enhancing memory, concentration, and longevity.

Modern pharmacological investigations have validated several of these traditional claims. *Bacopa monnieri* extracts have been shown to improve cognitive function, reduce anxiety, modulate the dopaminergic and serotonergic systems, protect neurons from oxidative stress, and demonstrate activity in animal models of Alzheimer's disease and epilepsy. Beyond neurology, the plant has demonstrated anti-inflammatory, antioxidant, hepatoprotective, immunomodulatory, antitumor, and adaptogenic properties, establishing it as a plant of broad therapeutic significance.



Fig.1.1 Bacopa Monnieri Plants

1.2 Bacosides: The Primary Bioactive Constituents

The pharmacological activity of *Bacopa monnieri* is primarily attributed to a class of triterpenoid saponins collectively termed "bacosides." Over 30 distinct bacosides have been identified, most being jujubogenin or pseudojujubogenin glycosides with dammarane-type triterpenoid aglycone cores. The two historically recognized fractions are Bacoside A and Bacoside B, first reported by Chatterji et al. in 1963.

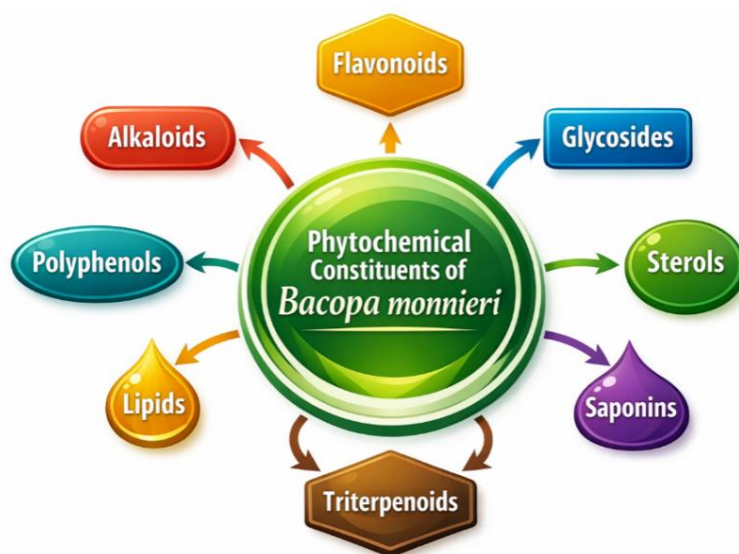


Fig1.2.1 Phytochemicals of Bacopa Monnieri

The major saponins isolated from *bacopa monnieri* include:

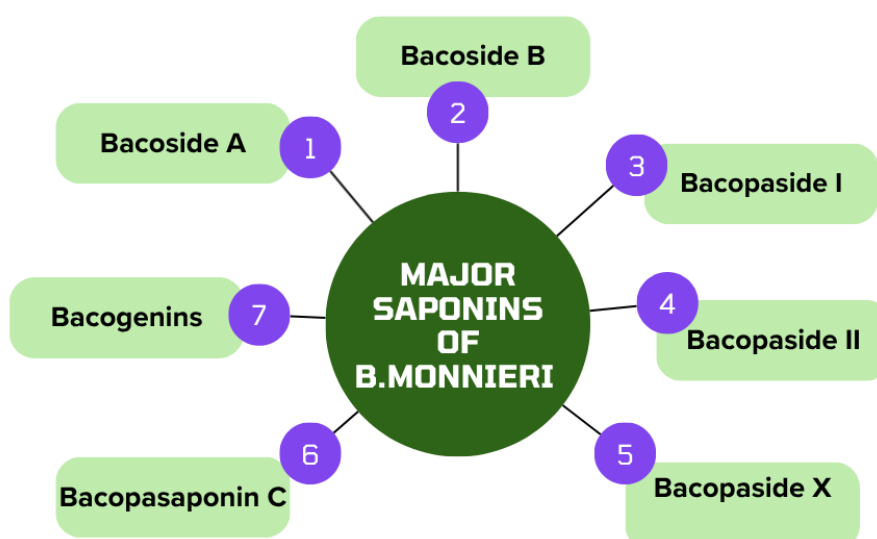


Fig.1.2.2 Major saponins of Bacopa Monnieri

1.2.1 Bacoside A

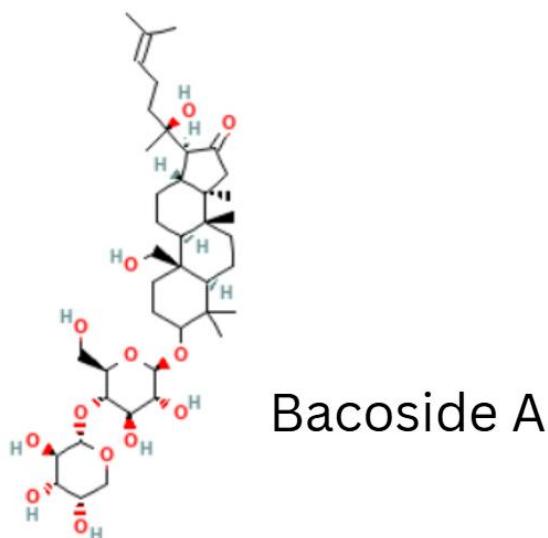


Fig.1.2.1.1 Structure of Bacoside A

Bacoside A is not a single compound but a mixture of several closely related triterpenoid saponins. It is considered the major active constituent of *Bacopa monnieri* and contributes significantly to the plant's medicinal properties.

Bacoside A is composed mainly of:

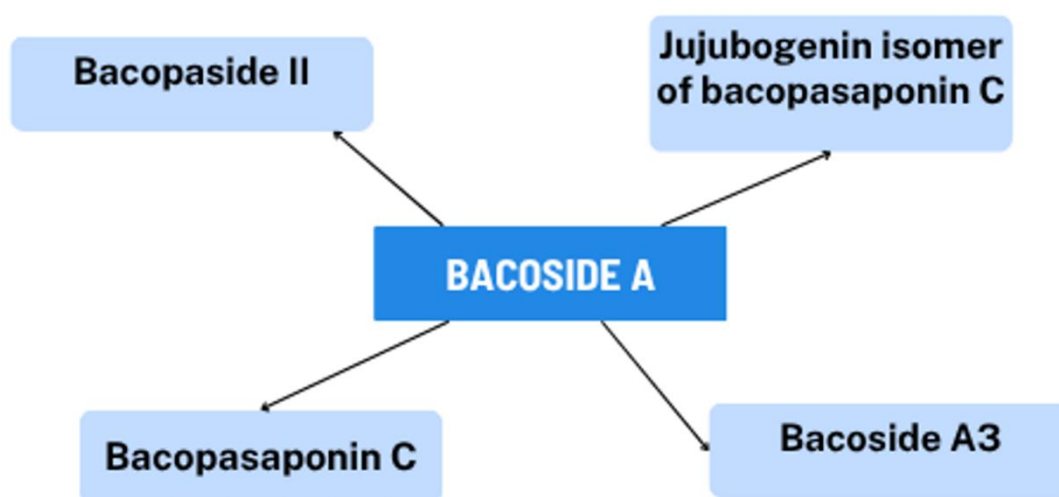


Fig.1.2.1.2 Composition of Bacoside A

Structurally, Bacoside A belongs to the class of dammarane-type triterpenoid glycosides. It contains sugar moieties attached to a triterpene backbone, which influences its biological activity, solubility, and pharmacokinetic properties.

Bacoside A has attracted major scientific attention because of its role in:

1. Enhancing memory and cognition
2. Improving synaptic transmission
3. Protecting neurons from oxidative stress
4. Increasing cerebral blood flow
5. Modulating neurotransmitters such as acetylcholine, serotonin, dopamine, and GABA
6. Reducing inflammation in the brain
7. Preventing neurodegeneration

Due to these effects, Bacoside A is being investigated for use in disorders such as Alzheimer's disease, Parkinson's disease, epilepsy, depression, anxiety, attention deficit hyperactivity disorder, and cognitive impairment.

1.2.2 Bacoside B

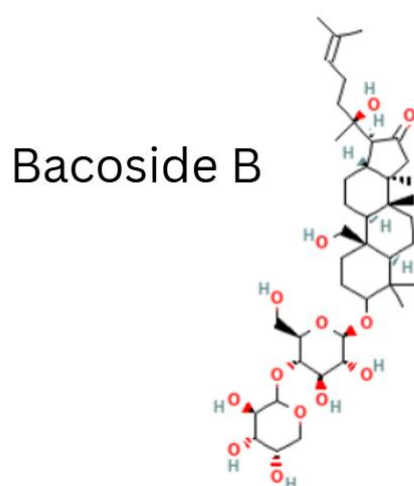


Fig.1.2.2.1 Structure of Bacoside B

Bacoside B is another important saponin fraction present in *Bacopa monnieri*. Similar to Bacoside A, it is also a mixture of related glycosidic compounds. Chemically, Bacoside B is often considered to be closely related to Bacoside A and differs mainly in stereochemical arrangement or optical activity.

Some studies suggest that Bacoside B may represent an artifact formed during the isolation process, while others consider it a naturally occurring compound group. Because of the structural similarities between Bacoside A and Bacoside B, both are believed to contribute to the pharmacological effects of *Bacopa monnieri*.

Bacoside B has been associated with:

1. Antioxidant activity
2. Neuroprotective action
3. Anti-inflammatory effects
4. Improvement in learning and memory
5. Cellular repair and regeneration
6. Modulation of cholinergic signaling

Although Bacoside A is generally regarded as the more potent and better-characterized fraction, Bacoside B is also important because it may act synergistically with Bacoside A to produce the overall therapeutic effects of *Bacopa monnieri*.

1.3 Need for Comparative Study and Research Gap

A comparative study between Bacoside A and Bacoside B is important because both compounds are major bioactive constituents of *Bacopa monnieri*, yet their exact chemical identities, pharmacological roles, and relative therapeutic significance are still not fully understood.

Many commercial *Bacopa* extracts are standardized according to total bacoside content, but differences in the proportion of Bacoside A and Bacoside B may influence biological activity. Therefore, understanding the similarities and differences between these two compounds is essential for:

- Proper standardization of herbal extracts
- Quality control of herbal formulations
- Development of effective nutraceuticals
- Drug discovery and development
- Determining which compound has greater therapeutic potential
- Identifying synergistic mechanisms between the two compounds

Comparative studies can also help determine whether one compound has superior antioxidant, anti-cancer, anti-inflammatory, or neuroprotective activity.

1.4 Why Bacoside B Has Not Been Studied Until Now

Bacoside B has been studied much less than Bacoside A mainly because:

1. Bacoside A is considered the major active fraction of *Bacopa monnieri* and most early pharmacological studies focused on the compound believed to be most responsible for memory enhancement and neuroprotection.
2. There is uncertainty about whether Bacoside B is actually a separate naturally occurring compound. Some researchers suggest it may be an artifact formed during extraction or isolation of Bacoside A due to stereochemical changes. Because of this confusion, many researchers preferred to work on Bacoside A, whose composition is better defined.
3. Bacoside B is chemically very similar to Bacoside A, so many papers simply discuss “total bacosides” instead of studying B separately.
4. Pure isolation of Bacoside B is more difficult. Since it is often present in lower amounts and mixed with other saponins, researchers may not obtain enough purified compounds for detailed biological testing.
5. Most commercial extracts are standardized mainly for total bacoside content or Bacoside A content, so research funding and industry attention have focused more on Bacoside A.
6. There are fewer molecular docking, pharmacokinetic, and in vitro studies specifically dedicated to Bacoside B, which creates a cycle: fewer studies lead to less available information, which leads future researchers to continue choosing Bacoside A instead.

1.5 Cancer as a Multi-Target Disease: Rationale for This Study

Cancer is a complex, multi-factorial disease characterized by dysregulation of numerous interconnected signaling pathways. Single-target therapeutic approaches often fail due to compensatory pathway activation and resistance emergence. *Bacopa monnieri*, as a complex phytochemical mixture, is an ideal candidate for network pharmacological investigation of multi-target anticancer effects. The present study focuses on two major cancers:

- Glioblastoma Multiforme (GBM) — an aggressive brain tumor with limited treatment options, characterized by high EGFR expression and dysregulated kinase signaling
- Hepatocellular Carcinoma (HCC/Liver Cancer) — the most common primary liver cancer, characterized by AKT1/PI3K and TP53 mutations

SECTION 2: METHODOLOGY

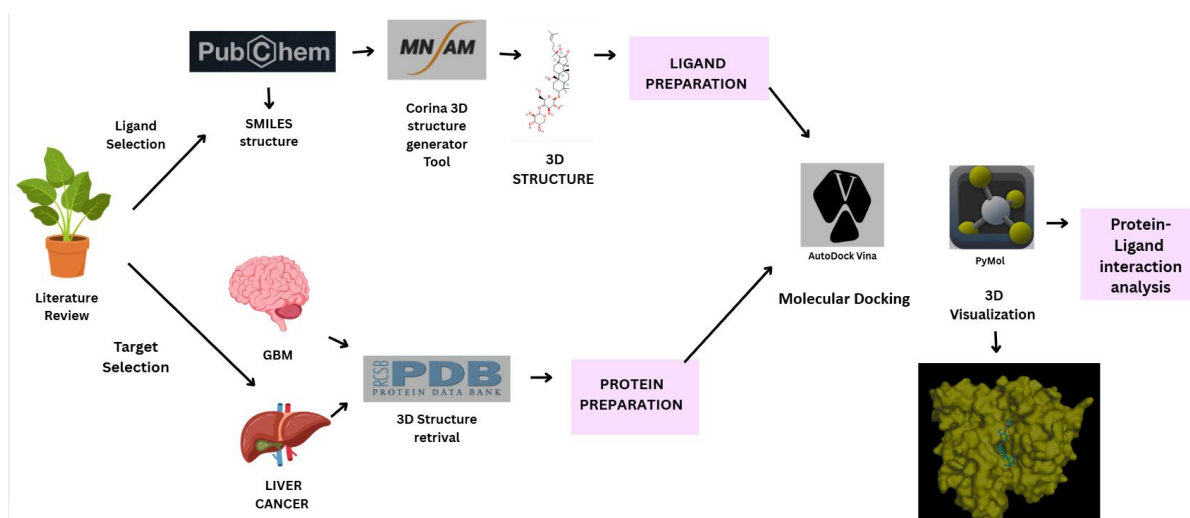


Fig.2.1 Methodology work flow

2.1 Literature Review and Target Selection

The study commenced with a detailed literature review to understand the key therapeutic targets involved in Glioblastoma Multiforme (GBM) and Hepatocellular Carcinoma (liver cancer). Based on the findings, specific protein targets were carefully selected for further analysis using network pharmacology approaches to identify cancer hub proteins central to oncogenic signaling pathways.

2.2 Ligand Preparation

Bioactive compounds Bacoside A (PubChem CID: 92043183) and Bacoside B were collected from the PubChem database in the form of SMILES structures. These 2D structures were then converted into 3D forms using the Corina tool. After that, the ligands were prepared and optimized to ensure proper geometry, correct charge distribution, and minimum energy state.

2.3 Target Protein Selection and Preparation

Three-dimensional structures of the selected proteins were obtained from the Protein Data Bank (RCSB PDB). These proteins were then prepared by: removing unwanted molecules (water, co-crystallized ligands), adding hydrogen atoms, assigning Kollman/AMBER charges, and fixing any structural issues to make them suitable for docking. Active site coordinates were

defined based on co-crystallized ligand positions or known catalytic residues. The following cancer hub proteins were selected:

Target Protein	PDB ID	Biological Role	Cancer Relevance	Cancer Type
AKT1	3O96	Serine/threonine kinase; cell survival, proliferation	PI3K/AKT/mTOR dysregulation; overexpressed in breast, ovarian, liver cancers	HCC
TP53	1TUP	Tumor suppressor transcription factor; apoptosis	Mutated in >50% of all human cancers	HCC
EGFR	1M17	Receptor tyrosine kinase; growth factor signaling	Overexpressed in GBM, lung, breast cancers	GBM
TNF	2AZ5	Pro-inflammatory cytokine; NF- κ B activation	Tumor microenvironment signaling	HCC
IL6	1P9M	Pleiotropic cytokine; STAT3 activation	Immune evasion in multiple cancers	HCC
VEGFA	3C7Q	Angiogenic growth factor	Tumor neovascularization	Both
HSP90AA1	3B28	Molecular chaperone; oncoprotein stabilization	Stabilizes HER2, AKT, CDK4, mutant proteins	Both
SRC	2BDJ	Non-receptor tyrosine kinase; invasion, metastasis	Invasion, EMT, cytoskeletal remodeling in GBM	GBM
CASP3	2H5I	Cysteine protease; apoptosis execution	Apoptosis resistance in cancer cells	Both

Table 1. Cancer hub proteins used in this study, their PDB IDs, biological roles, and cancer relevance.

2.4 Binding Site Prediction (P2Rank)

Probable ligand binding sites were predicted using P2Rank software, a machine learning-based tool used to predict ligand binding sites on protein structures. P2Rank scans protein surfaces to identify cavities and chemical environments, using a trained ML model to assign a pocket score and binding probability to each identified site. The score indicates pocket quality — the higher the score, the better the binding potential. Average conservation indicates the evolutionary importance of the pocket (how conserved the amino acids in that pocket are across different species).

2.5 Molecular Docking Protocol

Molecular docking was carried out using AutoDock Vina to study how well the ligands bind to the target proteins. Both ligands were docked into each protein's active site under identical parameters to ensure comparability. Binding affinities are reported in kcal/mol, where more negative values indicate stronger predicted binding affinity. Multiple docking poses were generated for each complex, and the lowest-energy pose was selected for interaction analysis.

2.6 Visualization and Interaction Analysis

2.6.1 Pymol

Protein-ligand complexes were visualized using PyMOL to observe three-dimensional binding orientations and important interactions such as hydrogen bonds and hydrophobic contacts.

2.6.2 Interaction Map Analysis: BIOVIA Discovery Studio:

Post-docking analysis was performed using BIOVIA Discovery Studio Visualizer. Two-dimensional (2D) protein-ligand interaction diagrams were generated for each of the 18 docking complexes (9 proteins $\times$ 2 ligands). These maps provide a comprehensive visual summary of all non-covalent interactions between the ligand and protein residues, categorized as follows:

- Hydrogen bonds: Directional interactions between hydrogen bond donors and acceptors (shown in green in Discovery Studio).
- Hydrophobic interactions: Van der Waals contacts with non-polar residues lining hydrophobic pockets.
- Pi-pi stacking: Face-to-face or edge-to-face interactions between aromatic systems.
- Pi-cation interactions: Between aromatic pi-systems and positively charged residues.
- Pi-alkyl interactions: Between aromatic rings and alkyl side chains.
- Electrostatic interactions: Charge-charge interactions with ionic residues.
- Solvent exposure: Regions of the ligand exposed to solvent without direct protein contacts.

2.7 Molecular Dynamics Simulation and ADMET Analysis (Planned)

Molecular dynamics (MD) simulations are planned to be performed to check the stability of the docked complexes over time under conditions similar to the human body. ADMET analysis using SwissADME is also planned to assess drug-likeness, pharmacokinetics, and toxicity properties of the ligands.

SECTION 3: RESULTS AND DISCUSSION

3.1 Docking Analysis Overview

Docking analysis of Bacoside A and Bacoside B was performed with two major cancers: Glioblastoma Multiforme (GBM) and Hepatocellular Carcinoma (liver cancer). Using P2Rank software, probable binding sites of cancer protein targets were predicted. Visualization of the 3D structure of ligand-protein complexes was performed using PyMOL and BIOVIA Discovery Studio.

3.2 Glioblastoma Multiforme (GBM)

Docking results of GBM with Bacoside A and B shown that GBM has more stronger binding affinity scores for Bacoside A with highest score of (-9.9) with (1IVO) crystal structure of human epidermal growth factor , represents the tyrosine kinase domain of the Epidermal Growth Factor Receptor (EGFR) . Binding affinity scores of GBM targets with Bacoside B had shown comparable but slightly weaker affinity scores. Bacoside A has shown consistently stronger binding (≤ -9 kcal/mol) were observed with targets like 1IVO, 5A63, 1GNG, 1CTO and better interaction across multiple GBM proteins.

Target Protein (PDB ID)	Bacoside A (kcal/mol)	Bacoside B (kcal/mol)	Better Binder
1IVO	-9.9	-8.6	Bacoside A
Jagged-1	-8.6	-8.1	Bacoside A
VEGFR	-8.3	-7.3	Bacoside A
5A63	-8.9	-8.2	Bacoside A
1GNG	-8.9	-9.4	Bacoside B
2YZQ	-7.8	-6.9	Bacoside A
1YC5	-7.8	-7.2	Bacoside A
3IQJ	-8.8	-8	Bacoside A
4XBM	-7.6	-7.1	Bacoside A
1CTQ	-9.5	-7.4	Bacoside A
2MH3	-7.9	-8	Same
2DB7	-6.5	-5.6	Bacoside A
2Z6H	-7.2	-6.8	Bacoside A
6BD4	-7.7	-6.3	Bacoside A

Table 2. Binding energies of Bacoside A and B for GBM.

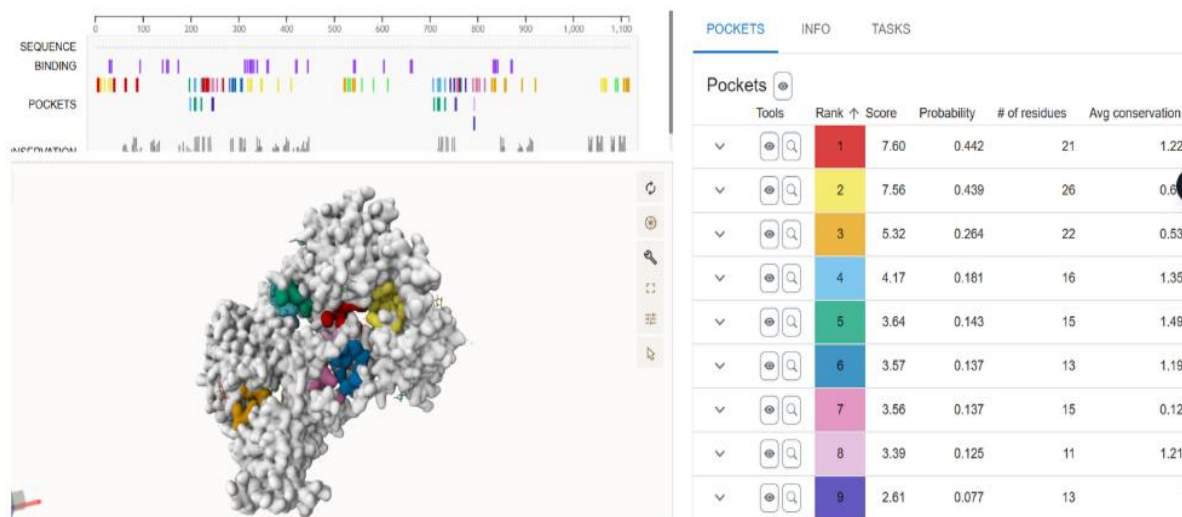


Fig.3.2.1 Probable pocket binding sites(1IVO) using P2Rank software

These are probable pocket binding sites of 1IVO, coloured regions are various pocket binding sites each colour has different pocket locations (cavities/grooves) where ligands can bind. The top bar shows sequence and binding regions. The score indicates pocket quality, higher the score better binding potential. Average conservation is the evolutionary importance of the pocket i.e (how conserved (unchanged) the amino acids in that pocket are across different species). P2Rank analysis identified multiple potential binding pockets on the 1IVO structure, with the top-ranked pocket showing a score of 7.60 and a probability of 0.442, indicating a favourable binding region.

Binding affinity scores and Pymol visualization of Bacoside A and B with Liver cancer targets:

Protein: 1IVO

```
done.
Refining results ... done.
```

mode	affinity (kcal/mol)	dist from best mode rmsd l.b. rmsd u.b.
1	-9.9	0.000 0.000
2	-8.9	2.350 5.016
3	-8.9	1.787 5.040
4	-8.8	4.426 7.585
5	-8.4	1.916 4.325
6	-8.4	3.635 10.122
7	-8.3	4.530 11.681
8	-8.2	4.972 10.546
9	-8.2	2.470 4.816

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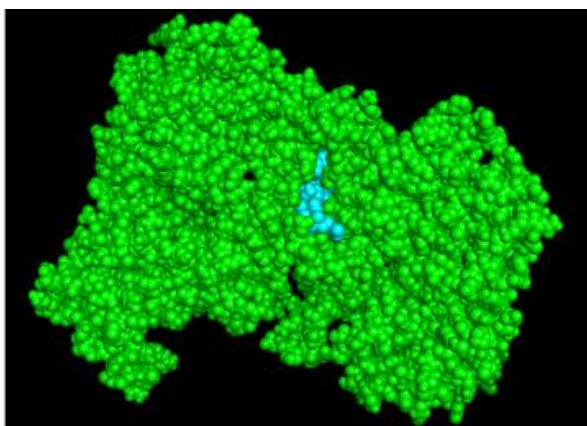
Fig: Binding affinity for Bacoside A

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done.
Refining results ... done.
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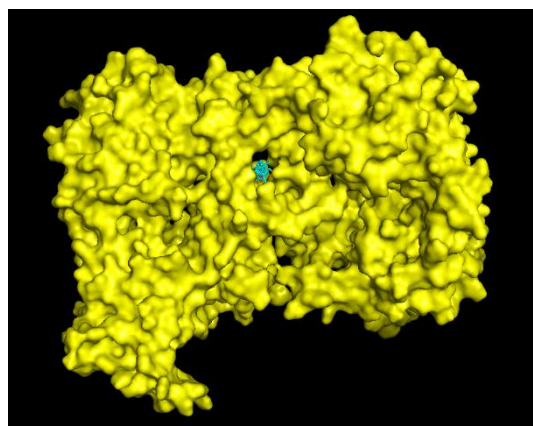
mode	affinity (kcal/mol)	dist from best mode rmsd l.b. rmsd u.b.
1	-8.6	0.000 0.000
2	-8.5	4.695 13.005
3	-8.2	3.666 14.028
4	-8.1	18.791 23.159
5	-8.1	3.068 5.034
6	-7.8	2.929 5.903
7	-7.6	7.427 14.687
8	-7.5	7.317 14.106
9	-7.5	3.635 13.563

```
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

Protein : 4X17

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode |   affinity | dist from best mode
   | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.6      0.000      0.000
2      -8.5     15.185     20.046
3      -8.5     10.914     14.854
4      -8.5     13.975     18.304
5      -8.5      7.531     12.728
6      -8.4      7.580     13.762
7      -8.4     17.093     20.837
8      -8.4     20.900     23.960
9      -8.4     17.215     20.463

Writing output ... done.
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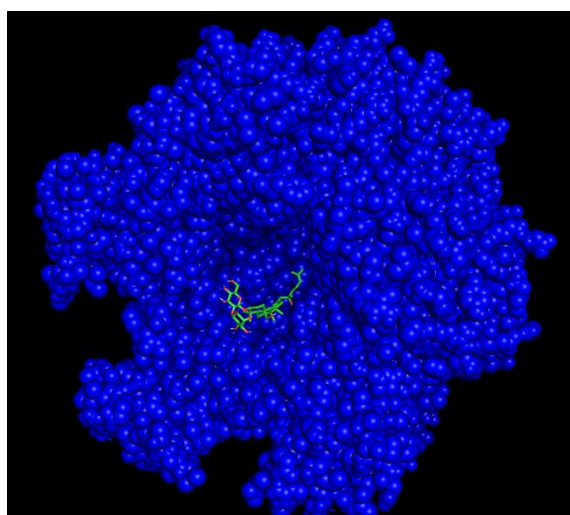
Fig: Binding affinity for Bacoside A

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

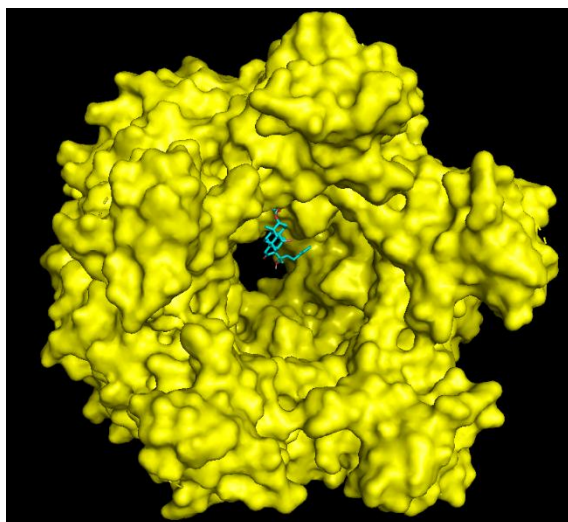
mode |   affinity | dist from best mode
   | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.1      0.000      0.000
2      -8.0     11.272     17.029
3      -8.0      8.261     10.445
4      -7.9      9.464     11.289
5      -7.8     16.066     18.987
6      -7.7     11.459     15.891
7      -7.7      8.796     13.921
8      -7.6     17.727     19.622
9      -7.5     19.622     22.294

Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

Protein: 1YWN

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

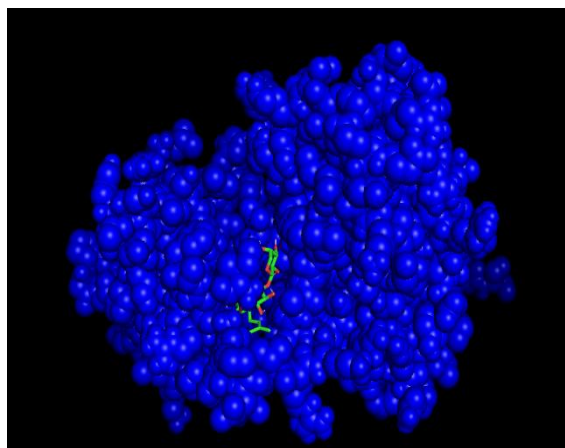
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.3      0.000      0.000
2      -8.1      3.245      5.947
3      -7.5      6.316     11.236
4      -7.3      1.783      2.737
5      -7.2      2.908      5.457
6      -7.2      3.632      6.144
7      -7.1      2.551     10.407
8      -7.1      2.612      5.249
9      -7.1      2.397     10.673
Writing output ... done.
```

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

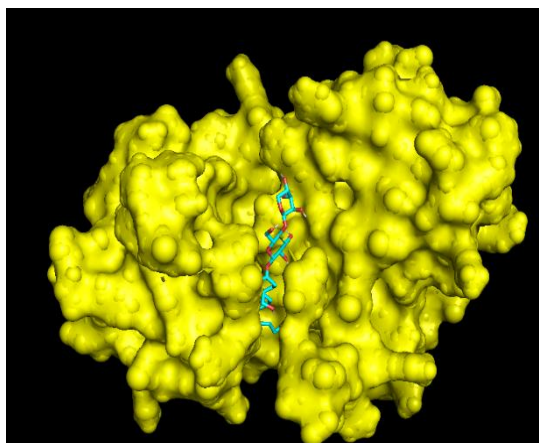
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -7.3      0.000      0.000
2      -6.9      3.106      4.526
3      -6.7      3.085      4.478
4      -6.6      6.644     11.311
5      -6.6      3.910     12.784
6      -6.5      3.975     12.880
7      -6.4      2.546      5.777
8      -6.2      7.030     13.016
9      -6.1      4.386     12.558
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

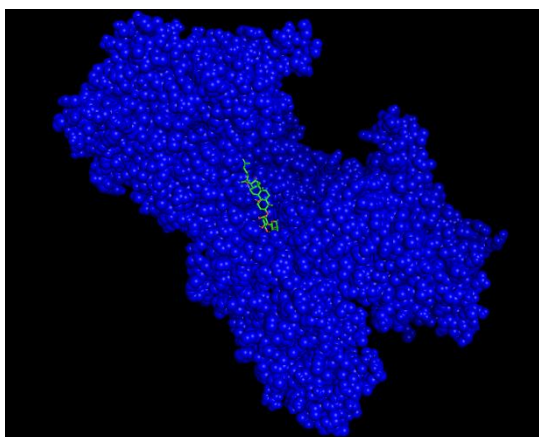
Protein: 5A63

```
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.9      0.000      0.000
2      -8.7     10.068     19.398
3      -8.4     13.538     18.489
4      -8.3     18.372     21.610
5      -8.0      7.737     16.931
6      -8.0     17.066     22.317
7      -7.9      6.067     16.616
8      -7.9     14.438     20.294
9      -7.8      8.063     14.081
Writing output ... done.
```

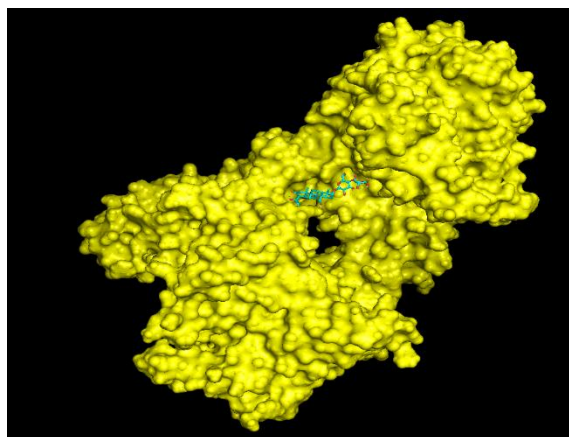
Fig: Binding affinity for Bacoside A

```
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.2      0.000      0.000
2      -8.1      6.703     11.588
3      -7.9      7.867     14.537
4      -7.8      7.698     12.752
5      -7.8      8.483     13.837
6      -7.8      1.298      1.678
7      -7.5     14.220     19.984
8      -7.5      2.086      3.729
9      -7.3      6.902     11.643
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

Protein : 1GNG

```

Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.9      0.000      0.000
2      -8.7     22.009     25.486
3      -8.6      6.276     10.794
4      -8.6     10.989     14.633
5      -8.5     11.024     14.905
6      -8.5     21.930     25.409
7      -8.5     21.936     25.657
8      -8.4      6.807     11.133
9      -8.2      1.215      3.801
Writing output ... done.

```

Fig: Binding affinity for Bacoside A

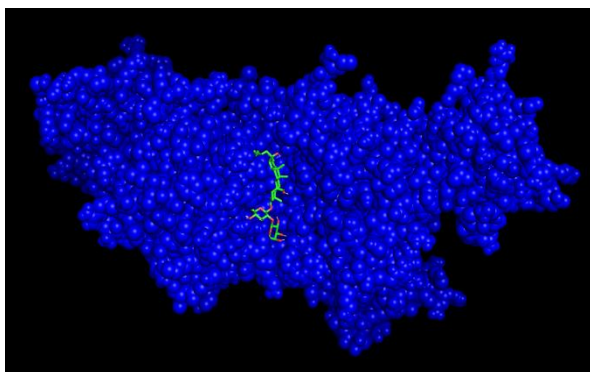
```

Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

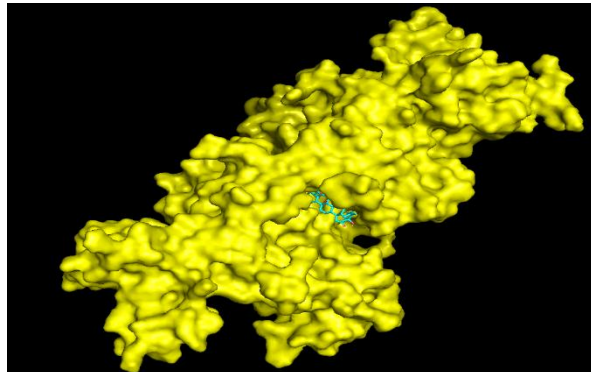
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -9.4      0.000      0.000
2      -9.4      4.375      9.460
3      -9.3      4.507      9.503
4      -9.3      2.355      4.935
5      -9.1      2.354      4.774
6      -9.0      8.260     19.608
7      -8.9      2.189      4.210
8      -8.9      2.933      5.261
9      -8.8     17.289     22.408
Writing output ... done.
C:\Users\pranj\OneDrive\Desktop\Docking 5>

```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



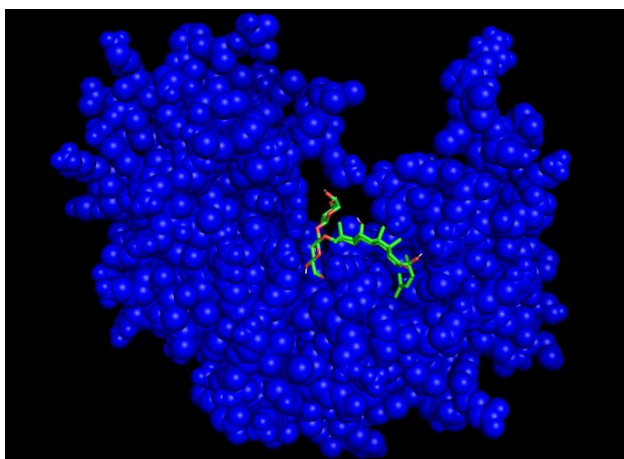
Docking site for Bacoside B

Protein: 2YZO

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -7.8 | 0.000 | 0.000
2 | -7.6 | 1.688 | 4.600
3 | -7.4 | 1.674 | 2.317
4 | -7.3 | 11.885 | 15.533
5 | -7.2 | 11.292 | 14.712
6 | -7.2 | 12.136 | 15.878
7 | -7.2 | 11.827 | 15.033
8 | -7.2 | 11.118 | 14.487
9 | -7.0 | 2.280 | 3.341
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

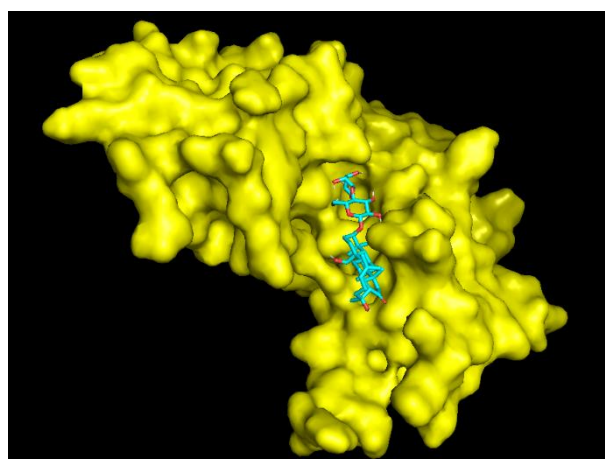


Docking site for Bacoside A

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -6.9 | 0.000 | 0.000
2 | -6.8 | 15.071 | 17.791
3 | -6.8 | 18.522 | 22.675
4 | -6.3 | 22.699 | 25.889
5 | -6.2 | 14.445 | 18.086
6 | -6.2 | 3.453 | 6.133
7 | -6.0 | 2.265 | 4.252
8 | -5.8 | 15.440 | 19.882
9 | -5.8 | 12.471 | 15.795
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside B

Protein: 1YC5

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

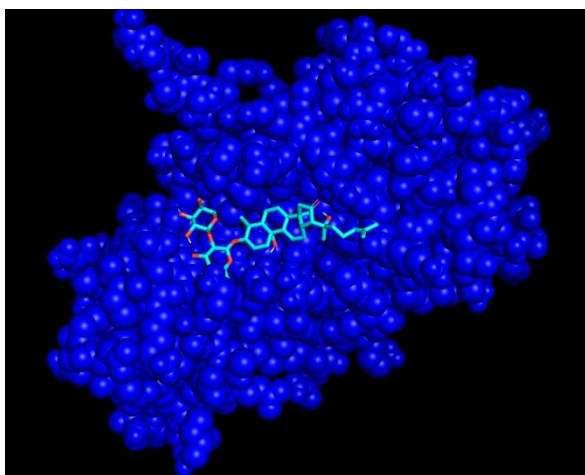
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -7.8 | 0.000 | 0.000
2 | -7.7 | 1.300 | 2.068
3 | -7.7 | 2.420 | 3.542
4 | -7.7 | 2.656 | 3.587
5 | -7.6 | 2.334 | 3.989
6 | -7.4 | 3.197 | 6.003
7 | -7.3 | 3.209 | 5.966
8 | -7.3 | 20.393 | 24.357
9 | -7.2 | 2.064 | 2.818
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

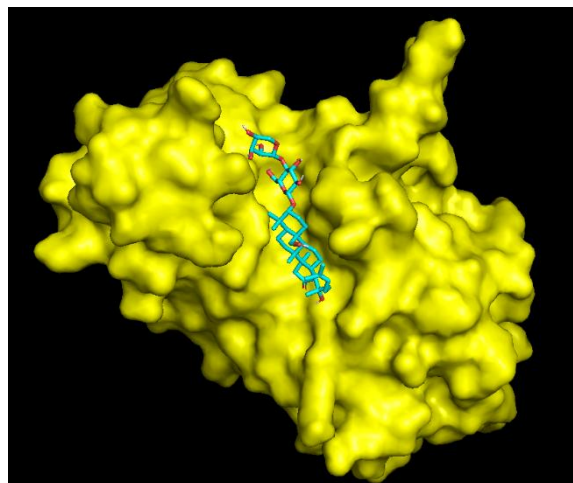
```
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -7.2 | 0.000 | 0.000
2 | -6.9 | 2.258 | 3.806
3 | -6.9 | 1.452 | 2.200
4 | -6.8 | 7.404 | 18.208
5 | -6.6 | 1.721 | 2.866
6 | -6.5 | 8.454 | 18.807
7 | -6.4 | 2.014 | 3.127
8 | -6.4 | 4.811 | 14.994
9 | -6.4 | 2.271 | 3.263
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

Protein: 3IQJ

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

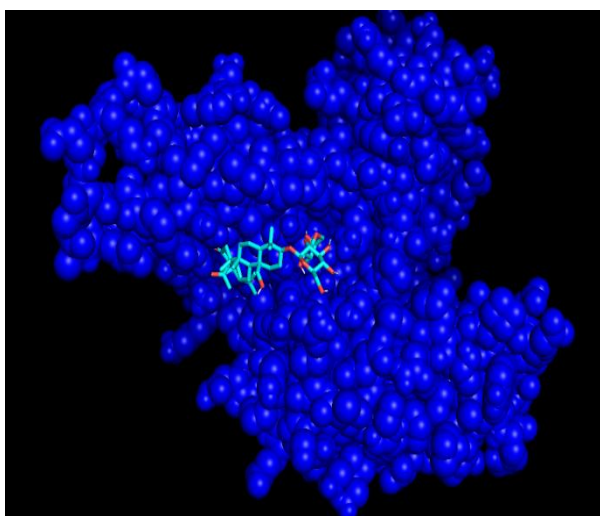
mode | affinity | dist from best mode
      | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.8      0.000      0.000
2      -8.6      3.715     10.415
3      -8.4      3.497     10.533
4      -8.1      1.462      4.140
5      -8.1      7.702     15.488
6      -8.0      8.434     16.180
7      -7.9      7.608     15.576
8      -7.6      4.313     11.124
9      -7.2     22.598     29.191
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

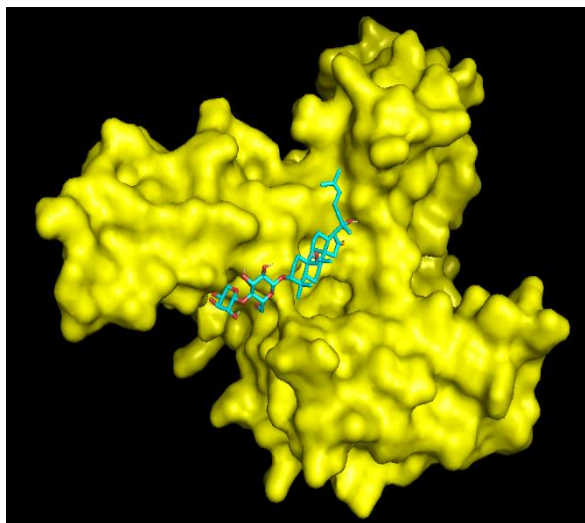
```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
      | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -8.0      0.000      0.000
2      -7.7      7.215     10.983
3      -7.3      5.612     13.051
4      -7.2      5.584      9.944
5      -7.1      7.222     12.435
6      -7.0      7.254     12.939
7      -7.0      3.153     13.035
8      -6.9      3.551     13.542
9      -6.9      5.556     13.081
Writing output ... done.
C:\Users\pranj\OneDrive\Desktop\docking 8 Raf>
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



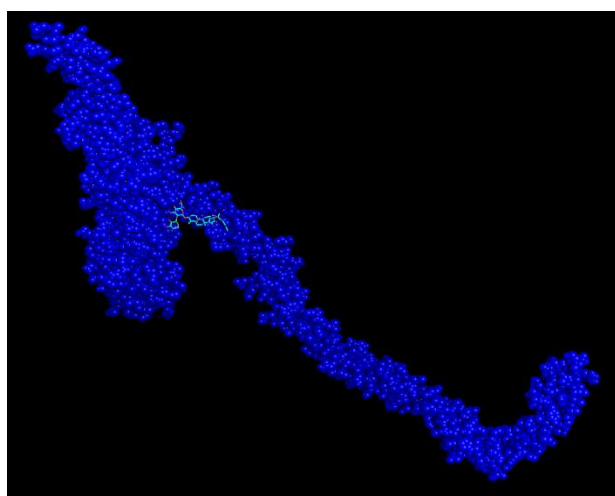
Docking site for Bacoside B

Protein: 4XBM

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -7.6      0.000      0.000
2      -7.5      8.850     14.572
3      -7.4      7.392     14.067
4      -7.3      5.142      8.030
5      -7.3      1.929      3.346
6      -7.2      7.503     14.814
7      -7.1      4.720     10.222
8      -7.1      1.996      4.338
9      -7.1      5.192     10.143
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

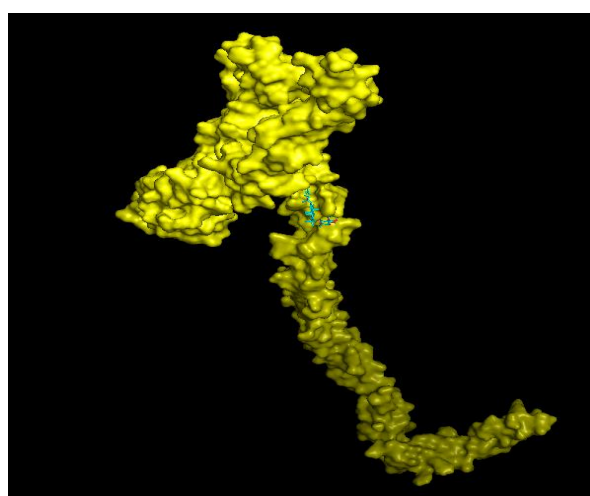


Docking site for Bacoside A

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -7.1      0.000      0.000
2      -7.0      8.919     13.453
3      -6.9     19.669     23.767
4      -6.8      8.421     14.635
5      -6.8     16.751     21.877
6      -6.6     17.443     22.032
7      -6.6     13.719     17.079
8      -6.5      4.714     11.636
9      -6.5     16.133     23.648
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside B

Protein: 1CTQ

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

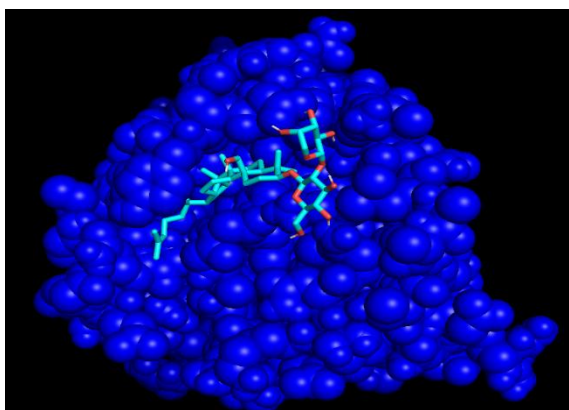
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -9.5      0.000      0.000
2      -9.0      1.700      2.111
3      -7.6      4.845      8.978
4      -7.1      4.349      7.493
5      -7.1     20.117     23.392
6      -7.1      5.227      9.254
7      -7.0     23.097     27.304
8      -6.9     18.085     22.361
9      -6.7     22.252     25.503
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

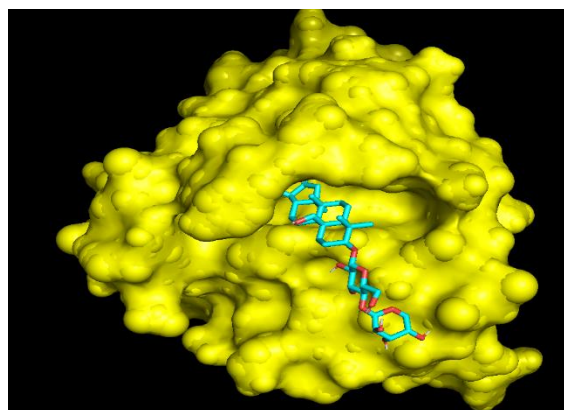
```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -7.4      0.000      0.000
2      -7.3     20.162     22.429
3      -7.3     20.362     22.457
4      -6.9      1.536      2.104
5      -6.8     21.634     26.608
6      -6.8     18.033     22.469
7      -6.7     22.024     25.637
8      -6.4     19.417     24.252
9      -6.3      4.441      9.602
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

Protein: 2MH3

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

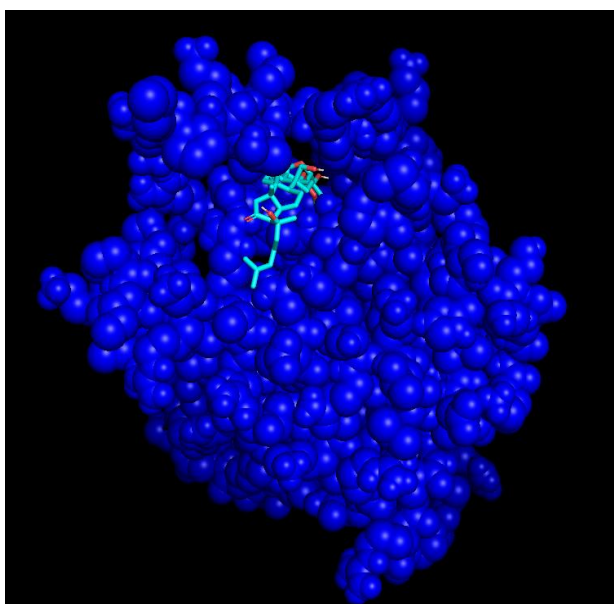
mode | affinity | dist from best mode
    | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -8.0 | 0.000 | 0.000
2 | -8.0 | 1.416 | 2.145
3 | -7.9 | 2.266 | 11.304
4 | -7.9 | 3.931 | 12.322
5 | -7.7 | 2.267 | 11.108
6 | -7.6 | 2.454 | 11.604
7 | -7.5 | 1.963 | 3.049
8 | -7.4 | 2.605 | 11.991
9 | -7.4 | 1.517 | 2.091
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

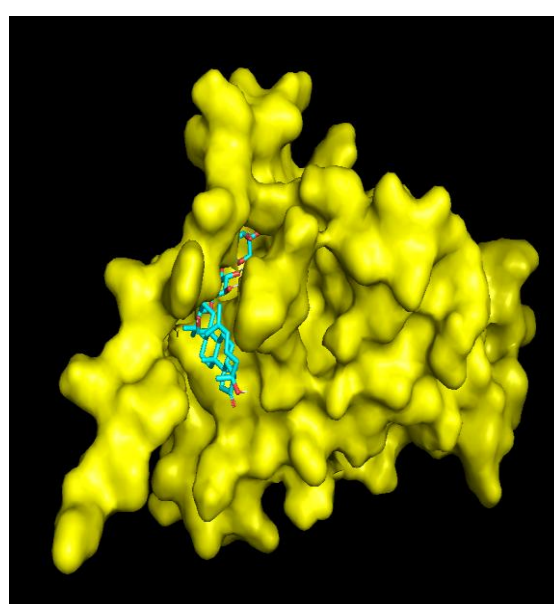
```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
    | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -7.9 | 0.000 | 0.000
2 | -7.8 | 1.281 | 1.709
3 | -7.4 | 1.746 | 2.284
4 | -7.4 | 3.542 | 12.229
5 | -7.0 | 13.307 | 21.060
6 | -6.9 | 12.757 | 19.729
7 | -6.6 | 5.217 | 14.453
8 | -6.4 | 3.584 | 6.013
9 | -6.4 | 1.921 | 3.231
Writing output ... done.
```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

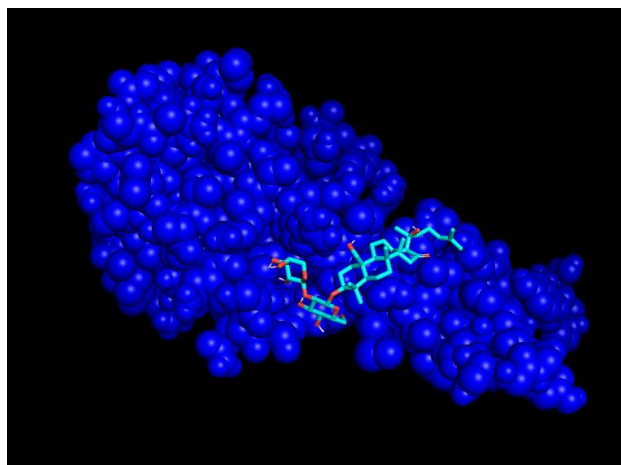
```

Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -6.5 | 0.000 | 0.000
2 | -6.2 | 1.216 | 1.807
3 | -6.1 | 1.572 | 4.679
4 | -6.0 | 16.574 | 20.658
5 | -5.9 | 0.907 | 4.101
6 | -5.9 | 2.001 | 3.329
7 | -5.6 | 1.568 | 2.194
8 | -5.5 | 17.913 | 22.245
9 | -5.4 | 7.978 | 13.855
Writing output ... done.

```

Fig: Binding affinity for Bacoside A



Docking site for Bacoside A

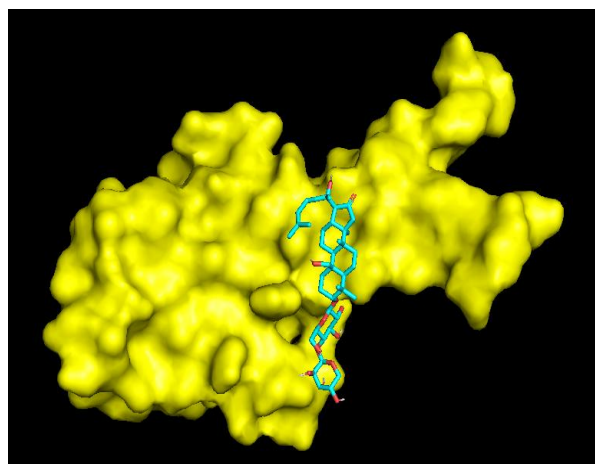
```

Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -5.6 | 0.000 | 0.000
2 | -5.4 | 1.545 | 2.102
3 | -5.1 | 21.063 | 23.751
4 | -5.1 | 25.493 | 27.646
5 | -5.1 | 16.491 | 21.418
6 | -5.0 | 2.079 | 2.993
7 | -4.9 | 9.860 | 13.500
8 | -4.9 | 17.569 | 21.619
9 | -4.9 | 20.698 | 24.770
Writing output ... done.

```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside B

Protein : 2Z6H

```

Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -7.2 | 0.000 | 0.000
2 | -7.0 | 22.518 | 25.738
3 | -7.0 | 2.733 | 4.208
4 | -7.0 | 7.577 | 13.034
5 | -6.9 | 1.670 | 2.900
6 | -6.9 | 13.456 | 21.162
7 | -6.8 | 3.990 | 6.643
8 | -6.8 | 6.992 | 12.420
9 | -6.7 | 15.462 | 21.667
Writing output ... done.

```

Fig: Binding affinity for Bacoside A

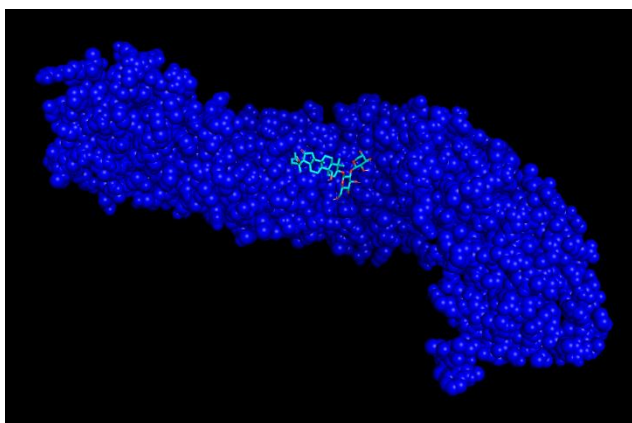
```

Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

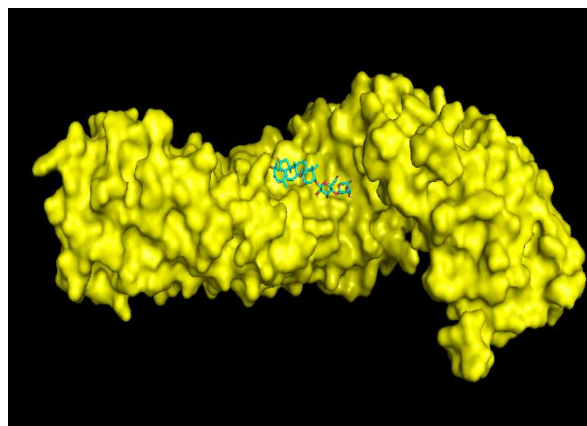
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1 | -6.8 | 0.000 | 0.000
2 | -6.7 | 1.293 | 1.986
3 | -6.4 | 6.141 | 12.409
4 | -5.8 | 8.600 | 14.469
5 | -5.6 | 7.118 | 11.268
6 | -5.6 | 7.373 | 11.523
7 | -5.5 | 7.871 | 13.772
8 | -5.4 | 7.044 | 11.152
9 | -5.4 | 3.937 | 5.174
Writing output ... done.

```

Fig: Binding affinity for Bacoside B



Docking site for Bacoside A



Docking site for Bacoside B

Protein : 6BD4

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

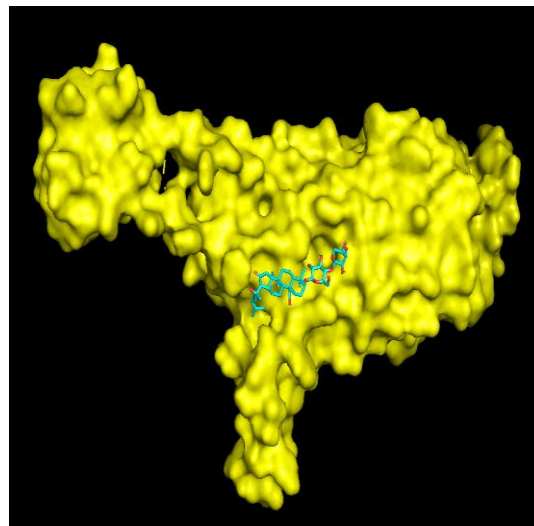
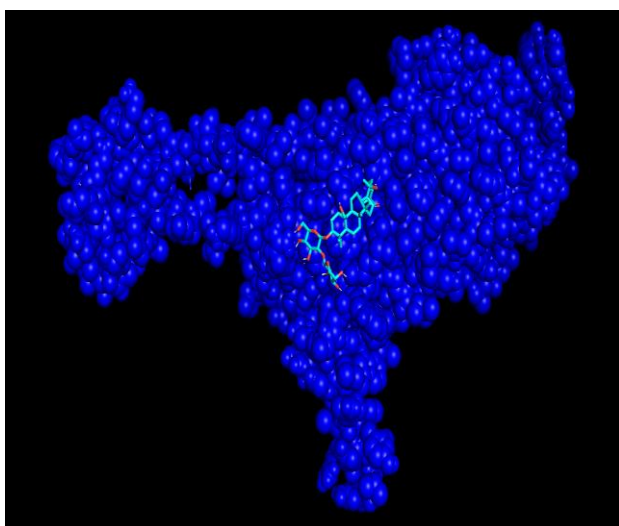
mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -7.7      0.000      0.000
2      -7.7      3.051     11.340
3      -7.5      3.046     11.427
4      -7.2      2.720     11.502
5      -7.1     20.173     26.917
6      -7.1     22.040     27.201
7      -7.0      2.635     10.918
8      -7.0     22.999     28.255
9      -7.0      2.376      3.967
Writing output ... done.
```

Fig: Binding affinity for Bacoside A

```
Performing search ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****
done.
Refining results ... done.

mode | affinity | dist from best mode
| (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
1      -6.3      0.000      0.000
2      -6.2      1.598      2.534
3      -6.0     25.122     30.126
4      -6.0     12.891     15.116
5      -5.9     13.025     15.077
6      -5.9      2.426      4.300
7      -5.9      7.564     12.488
8      -5.8     27.830     31.120
9      -5.7      7.289     10.663
Writing output ... done.
```

Fig: Binding affinity for Bacoside A



3.3 Protein Ligand interaction analysis using Discovery Studio

This section presents a detailed analysis of the docking results and 2D interaction maps for each of the nine cancer hub proteins. For each protein, the biological context is established, followed by individual analysis of Bacoside A and B binding, comparative interpretation, and a summary interaction table. The 2D interaction maps from BIOVIA Discovery Studio are presented alongside the analysis^{3.3}

1. TP53 / p53 tumor suppressor (DNA-binding protein): [PDB ID – 1IVO]

Bacoside A binds better to PDB ID 1IVO than Bacoside B. This is clearly shown by its lower docking score (−9.9 kcal/mol compared to −8.6 kcal/mol), which indicates a stronger and more stable interaction.)

Interaction parameter	Bacoside A	Bacoside B
Docking score	−9.9 kcal/mol	−8.6 kcal/mol ✓
Hydrogen bond	Multiple H-bonds (ARG321, GLU258, ASN211 region)	Multiple H-bonds (ASP238, HIS239, LEU240 region)
Hydrophobic interaction	Moderate hydrophobic interactions (ILE229, LEU regions)	Moderate hydrophobic interactions (PRO241, PRO257 region)
Pi-type interaction	Present (alkyl interactions with hydrophobic residues)	Present (alkyl interactions with PRO residues)
Overall assesment	Superior binding — best among the two	Strong and stable binding

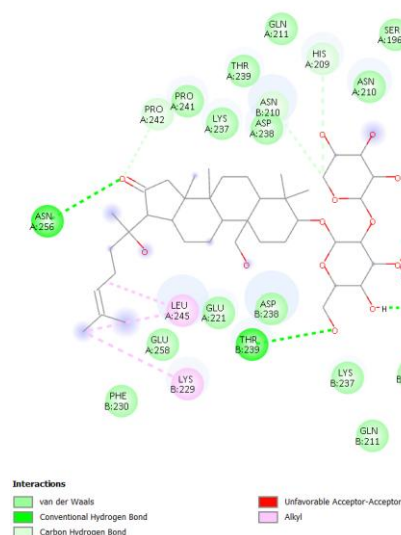


Fig : Bacoside A – 1IVO

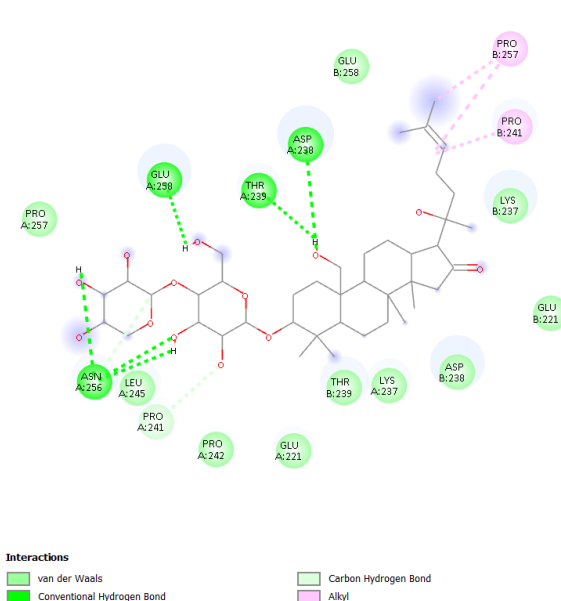


Fig : Bacoside B – 1IVO

Comparative Interpretation

Bacoside A forms multiple hydrogen bonds with residues such as ARG321 and GLU258, helping it stay firmly in the binding site. It also shows good hydrophobic interactions and spreads well across the binding pocket, which improves stability. Bacoside B also forms hydrogen bonds and hydrophobic interactions, but its overall interaction network is slightly less extensive. As a result, its binding is not as strong as Bacoside A.

2.JAG1 / Jagged1 (PDB: 4X17)

Bacoside A demonstrates the stronger binding affinity at PDB ID 4X17 (Jagged1 complex), with a docking score of -8.6 kcal/mol compared to -8.1 kcal/mol for Bacoside B, a difference of 0.5 kcal/mol that indicates a modest but potentially meaningful improvement in binding stability.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-8.6 kcal/mol	-8.1 kcal/mol ✓
Hydrogen bond	Multiple strong H-bonds with SER140, THR217, THR234	Fewer but key H-bonds with ARG276, HIS227
Hydrophobic interaction	Moderate engagement (VAL, PRO residues)	Extensive hydrophobic interactions (VAL222, VAL225, PRO223)
Pi-type interaction	Multiple, distributed	Few, localized
Overall assesment	Superior binding — best among the two	Strong and stable binding

Table 4. Interaction Comparison at JAG1 /(PDB: 4X17)

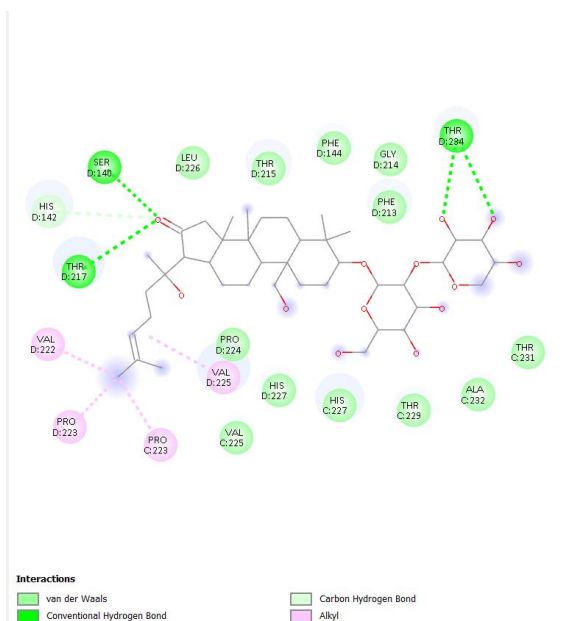


Fig : Bacoside A – 4X17

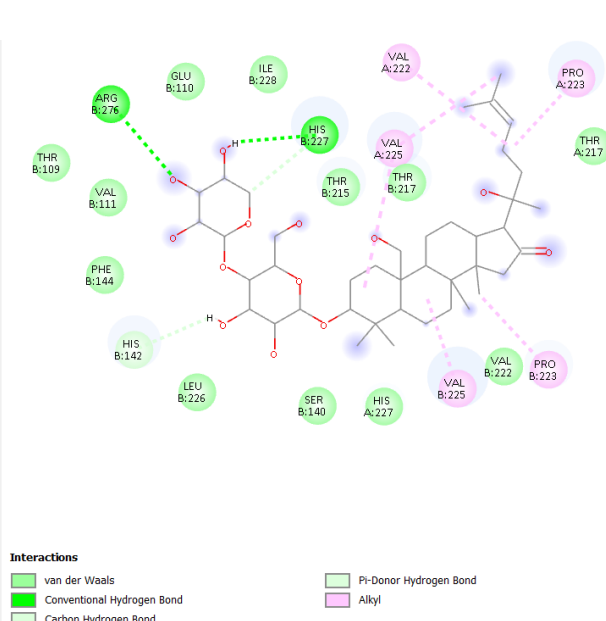


Fig : Bacoside B – 4x17

COMPARATIVE INTERPRETATION:

Bacoside B demonstrates a stronger binding affinity toward 4X17 compared to Bacoside A, as evidenced by its more favorable docking score (-8.6 kcal/mol) and a denser interaction network within the binding pocket. The enhanced hydrophobic contacts and additional pi-alkyl interactions enable Bacoside B to occupy the active site more effectively. Although Bacoside A forms multiple conventional hydrogen bonds, its overall interaction profile is less extensive than Bacoside B, indicating comparatively lower binding stability.

3.VEGFR2 (Receptor tyrosine kinase): [PDB ID -1YWN]

Bacoside A demonstrates a stronger binding affinity toward PDB ID 1YWN compared to Bacoside B, with a docking score of -8.3 kcal/mol versus -7.3 kcal/mol for Bacoside B. The 1.0 kcal/mol difference indicates a meaningful improvement in binding stability and suggests a more favorable interaction of Bacoside A within the binding pocket.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-8.3 kcal/mol ✓	-7.3 kcal/mol
Hydrogen bond	Multiple strong H-bonds (ASP1044, ASN1031, LYS1053, ILE1051) ✓	Fewer H-bonds (ASP1044, HIS1024)
Hydrophobic interaction	Moderate hydrophobic interactions (VAL846, LEU1033)	Limited hydrophobic contacts (LEU1033 region)

Pi-type interaction	Present but not dominant	Few, localized
Overall assesment	Strong and stable binding	Moderate binding

Table 5. Interaction Comparison at VEGFR2[PDB ID -1YWN]

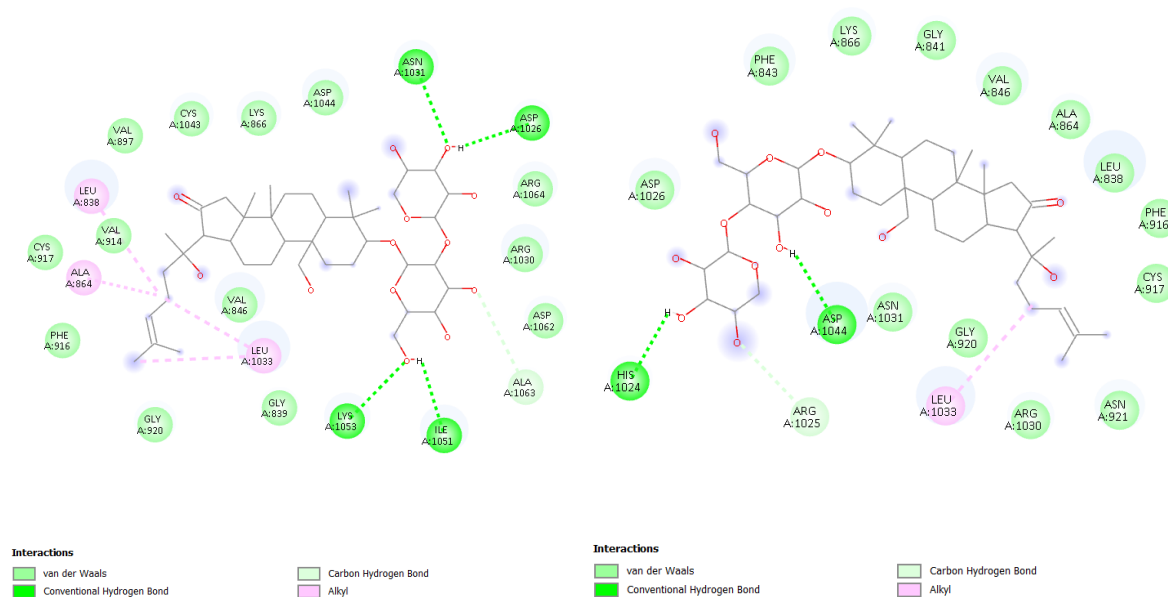


Fig : Bacoside A – 1YWN

Fig : Bacoside B – 1YWN

COMPARATIVE INTERPRETATION :

Bacoside A shows better binding with PDB ID 1YWN compared to Bacoside B, as seen from its lower docking score (-8.3 kcal/mol vs -7.3 kcal/mol). Bacoside A forms several strong hydrogen bonds with important residues like ASP1044, ASN1031, LYS1053, and ILE1051. These interactions help hold the ligand firmly in the active site. On the other hand, Bacoside B forms fewer hydrogen bonds and depends more on limited hydrophobic interactions, leading to weaker overall binding. Although Bacoside B does show some hydrophobic contacts, they are not enough to match the stability provided by the strong hydrogen bonding network of Bacoside A. As a result, Bacoside A has a more stable and effective interaction with the protein.

4. Gamma-secretase complex (Aspartyl protease / Hydrolase): [PDB ID – 5A63]

Bacoside A demonstrates a stronger binding affinity toward PDB ID 5A63 compared to Bacoside B, as indicated by its more favorable docking score (-8.9 kcal/mol) relative to -8.2

kcal/mol for Bacoside B. The 0.7 kcal/mol difference suggests a modest but meaningful improvement in binding stability.

Interaction parameter	Bacoside A	Bacoside B
Docking score	−8.9 kcal/mol ✓	−8.2kcal/mol
Hydrogen bond	Multiple strong H-bonds (PRO244, THR188, PHE184)	Fewer H-bonds (ASP185, ARG262)
Hydrophobic interaction	Moderate hydrophobic interactions (TRP289, ALA556, GLU550 region)	Limited hydrophobic contacts
Pi-type interaction	Presence of Pi–sigma and Pi–alkyl interactions	Minimal Pi interactions
Overall assesment	Strong and stable binding	Moderate binding

Table 6. Interaction Comparison at Gamma-secretase complex [PDB ID – 5A63]

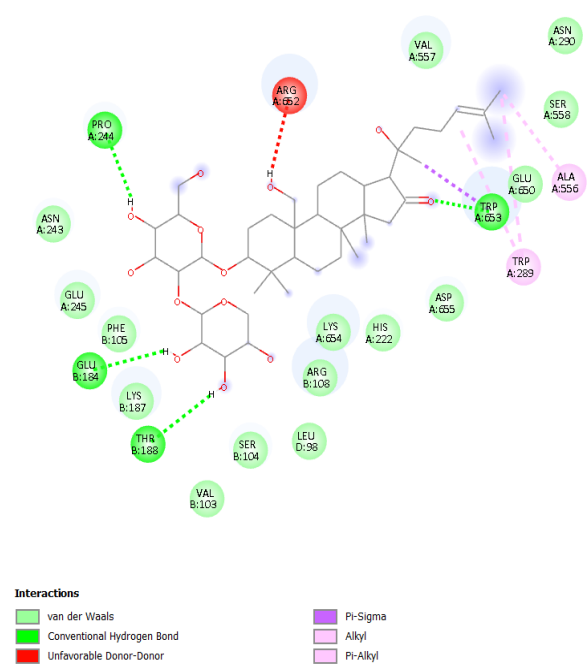


Fig : Bacoside A – 5A63

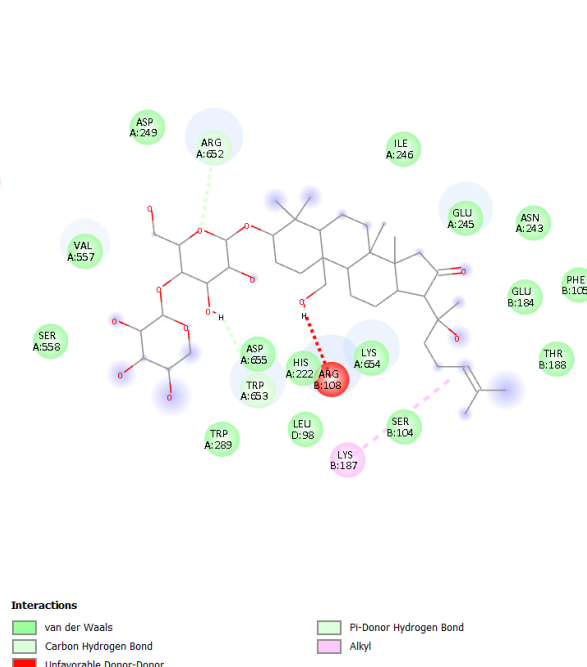


Fig : Bacoside B – 5A63

Comparative Interpretation:

Bacoside A demonstrates a stronger binding affinity toward PDB ID 5A63 compared to Bacoside B, as indicated by its more favorable docking score (−8.9 kcal/mol) relative to −8.2 kcal/mol for Bacoside B. Bacoside A forms multiple conventional hydrogen bonds with key residues such as PRO244, THR188, and PHE184, contributing significantly to stable anchoring within the binding pocket. Additionally, the presence of π -sigma and π -alkyl interactions enhances hydrophobic stabilization and improves ligand fitting. In contrast, Bacoside B exhibits fewer hydrogen bonding interactions and limited hydrophobic and π -type contacts, resulting in a less extensive interaction network. Consequently, Bacoside B shows comparatively weaker binding and reduced pocket stabilization relative to Bacoside A.

5. GSK3 β / Glycogen synthase kinase-3 beta (Serine/Threonine kinase): [PDB ID – 1GNG]

Bacoside B demonstrates a stronger binding affinity toward PDB ID 1GNG compared to Bacoside A, as evidenced by its more favorable docking score (−9.4 kcal/mol) relative to −8.9 kcal/mol for Bacoside A. The 0.5 kcal/mol difference suggests a modest but favorable improvement in binding stability.

Interaction parameter	Bacoside A	Bacoside B
Docking score	−8.9 kcal/mol ✓	−9.4kcal/mol
Hydrogen bond	Multiple strong H-bonds ((ARG309, SER310, THR309)	Fewer H-bonds (SER338, ASP341, ASN341)
Hydrophobic interaction	Moderate hydrophobic interactions (LEU321, ALA327 regionsregion)	Extensive hydrophobic interactions (LEU, ALA, ILE residues)
Pi-type interaction	Limited alkyl interactions	More pronounced Pi-alkyl interactions (PRO, HIS regions)
Overall assesment	Strong and stable binding	Superior binding — best among the two

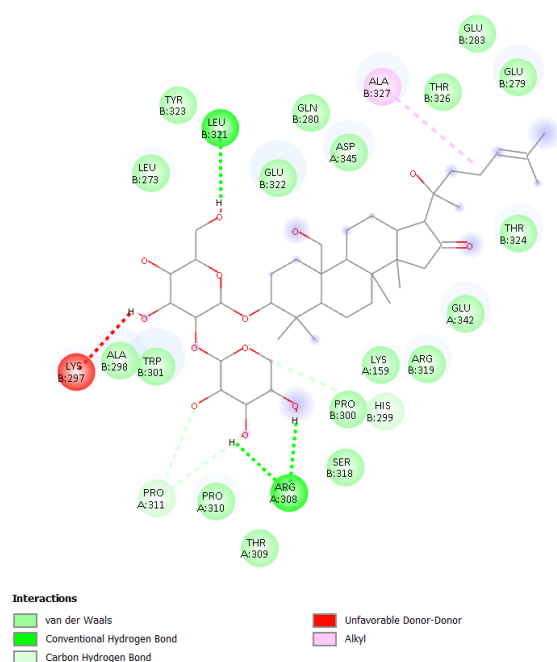


Fig : Bacoside A – 1GNG

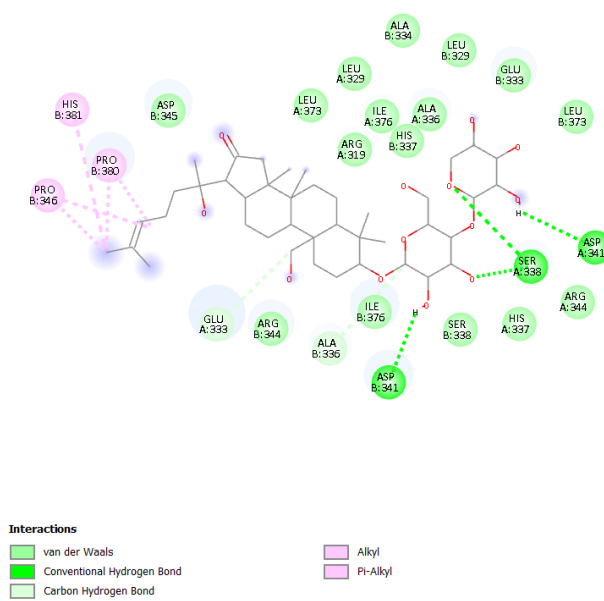


Fig : Bacoside B – 1GNG

Comparative Interpretation:

Bacoside B forms a stronger and more distributed interaction network within the binding pocket, including key hydrogen bonds with residues such as SER338, ASP341, and ASN341, along with extensive hydrophobic and π -alkyl interactions. These interactions allow Bacoside B to achieve better stabilization and deeper accommodation within the active site. In contrast, Bacoside A, although forming multiple hydrogen bonds, exhibits comparatively fewer hydrophobic and π -type interactions, resulting in a less extensive interaction network and slightly reduced binding stability.

6.Uncharacterized conserved protein (Unknown function / Structural genomics): [PDB ID – 2YZQ]

Bacoside A binds better to PDB ID 2YZQ than Bacoside B. This is clear from the docking scores, where Bacoside A has a lower (better) value of -7.8 kcal/mol compared to -6.9 kcal/mol for Bacoside B.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-7.8 kcal/mol ✓	-6.9 kcal/mol
Hydrogen bond	Multiple strong H-bonds (ARG33, THR176, THR180)	Fewer H-bonds (LEU152)
Hydrophobic interaction	Moderate hydrophobic interactions (ILE172, ALA252 regions)	Limited hydrophobic contacts

Pi-type interaction	Present (alkyl contacts contributing to stabilization)	Minimal Pi interactions
Overall assesment	Strong and stable binding,superior	weaker binding

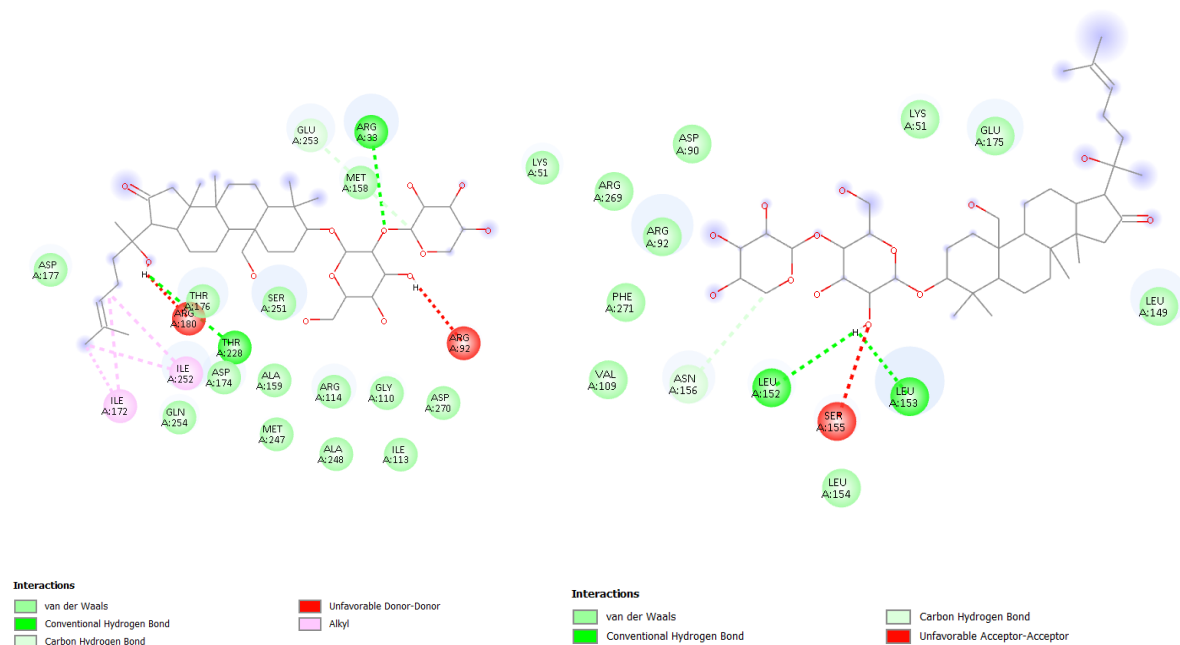


Fig : Bacoside A – 2YZQ

Fig : Bacoside B – 2YZQ

Comparative Interpretation

Bacoside A forms several strong hydrogen bonds with important residues like ARG33 and THR176, which help hold the molecule firmly in place. It also shows decent hydrophobic interactions, allowing it to sit more comfortably inside the binding pocket.

On the other hand, Bacoside B forms fewer hydrogen bonds and has weaker hydrophobic interactions. Because of this, it does not fit as well into the binding site and forms a less stable complex. Both molecules show some unfavorable interactions, but overall, Bacoside A still maintains better stability.

7.SIR2 / Sirtuin–p53 peptide complex (NAD-dependent deacetylase / Hydrolase): [PDB ID – 1YC5]

Bacoside A binds slightly better to PDB ID 1YC5 than Bacoside B. This is supported by its docking score of -7.8 kcal/mol compared to -7.2 kcal/mol for Bacoside B, indicating a more stable interaction.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-7.8 kcal/mol	-7.2 kcal/mol

Hydrogen bond	Multiple H-bonds	Strong H-bonds
Hydrophobic interaction	Moderate hydrophobic interactions (VAL, ILE, ALA residues)	Moderate hydrophobic interactions
Pi-type interaction	Limited alkyl interactions	Slightly better alkyl contacts
Overall assesment	Strong and stable binding — superior	Moderate binding

Table 9. Interaction Comparison at SIR2 [PDB ID – 1YC5]

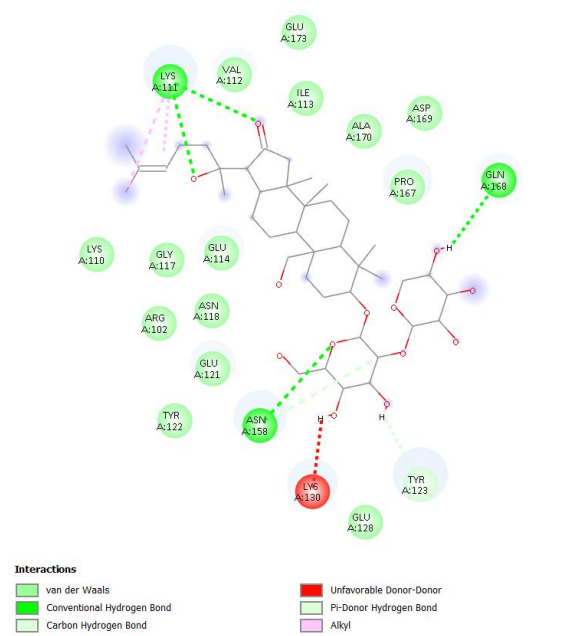


Fig : Bacoside A – 1YC5

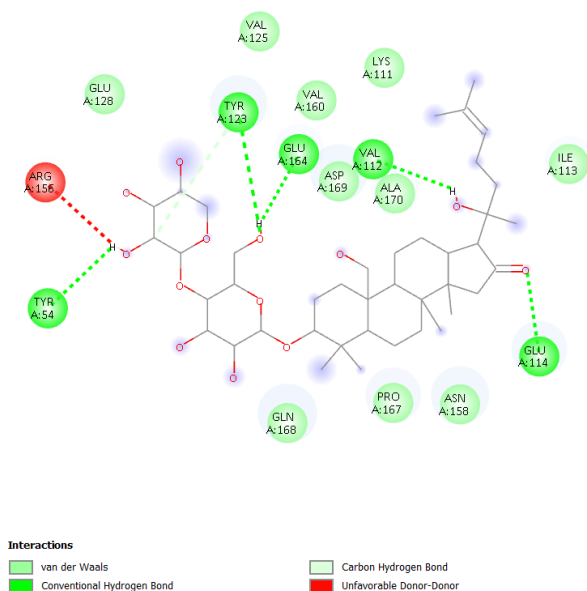


Fig : Bacoside B – 1YC5

Comparative Interpretation:

Bacoside A binds slightly better to PDB ID 1YC5 than Bacoside B. Bacoside A forms several hydrogen bonds with residues like LYS117, ASN115, and GLN124, which help stabilize the ligand in the binding site. It also shows decent hydrophobic interactions, allowing it to fit well within the pocket, although there are a few unfavorable interactions present. Bacoside B also forms good hydrogen bonds, especially with TYR123 and GLU114, and has slightly better alkyl (hydrophobic) contacts. However, its overall interaction network is not as strong or as well-balanced as Bacoside A.

8.(Sigma) protein in complex with Raf1 peptide (Signaling protein / Protein binding): [PDB ID – 3IQJ]

Bacoside A binds better to PDB ID 3IQJ than Bacoside B. This is supported by its lower docking score of -8.8 kcal/mol compared to -8.0 kcal/mol for Bacoside B, which indicates a more stable interaction.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-8.8 kcal/mol	-8.0 kcal/mol
Hydrogen bond	Multiple H-bonds (GLN57, SER63, ARG80)	Multiple H-bonds (ARG80, ASN50, LYS49)
Hydrophobic interaction	Moderate hydrophobic interactions (ILE183 region)	Moderate hydrophobic interactions (TRP51, VAL61 region)
Pi-type interaction	Limited alkyl interactions	Present (Pi-alkyl with TYR19)
Overall assesment	Strong and stable binding — superior	Moderate binding

Table 10. Interaction Comparison at [PDB ID – 3IQJ]

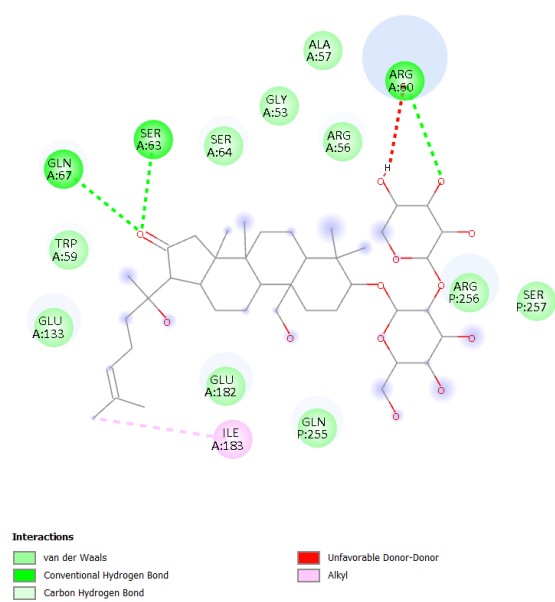


Fig : Bacoside A – 3IQJ

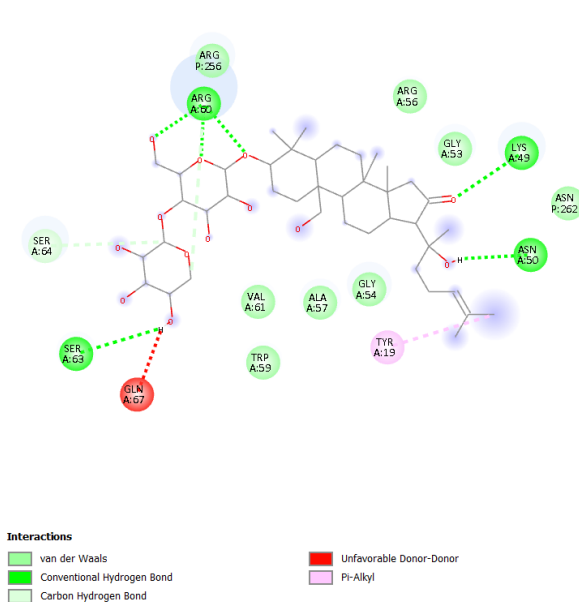


Fig : Bacoside B – 3IQJ

Comparative Interpretation:

Both ligands form multiple hydrogen bonds with the protein. However, Bacoside A shows a slightly better overall interaction pattern, with hydrogen bonds involving residues like GLN57, SER63, and ARG80, helping it stay well-positioned in the binding site.

Bacoside B also forms hydrogen bonds and shows some additional hydrophobic and π -type interactions, but its overall interaction network is not as well balanced as Bacoside A.

Although both ligands show a few unfavorable interactions, Bacoside A still achieves better stability inside the binding pocket.

9.DLL1 / Delta-like ligand 1 (Notch signaling ligand / Signaling protein): [PDB ID – 4XBM]

Bacoside A binds slightly better to PDB ID 4XBM than Bacoside B, as shown by its docking score of -7.6 kcal/mol compared to -7.1 kcal/mol for Bacoside B. This suggests that Bacoside A forms a somewhat more stable complex with the protein.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-7.6 kcal/mol	-7.1 kcal/mol
Hydrogen bond	Multiple H-bonds (GLN113, THR115, THR79)	Multiple H-bonds (ASP197, GLU222, THR222)
Hydrophobic interaction	Moderate hydrophobic interactions (PHE199, TYR183 regions)	Moderate hydrophobic interactions (PHE102, TYR220 regions)
Pi-type interaction	Present (Pi-sigma / Pi-alkyl with aromatic residues)	Present (Pi-alkyl interactions with PHE, TYR residues)
Overall assesment	Strong and stable binding — slightly superior	Moderate to strong binding

Table 11. Interaction Comparison at.DLL1[PDB ID – 4XBM]

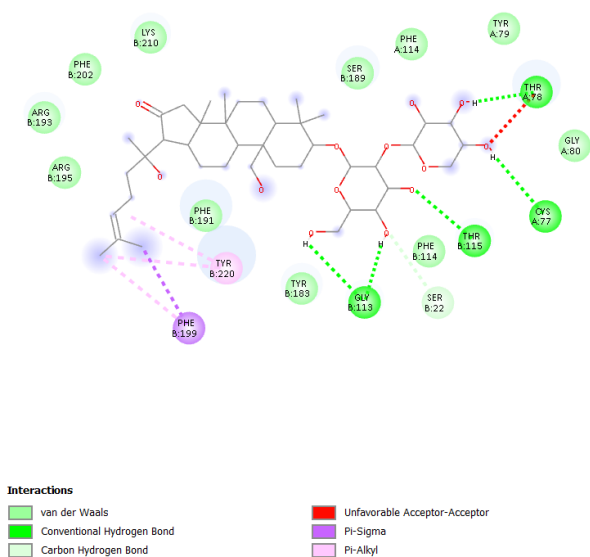


Fig : Bacoside A – 4XBM

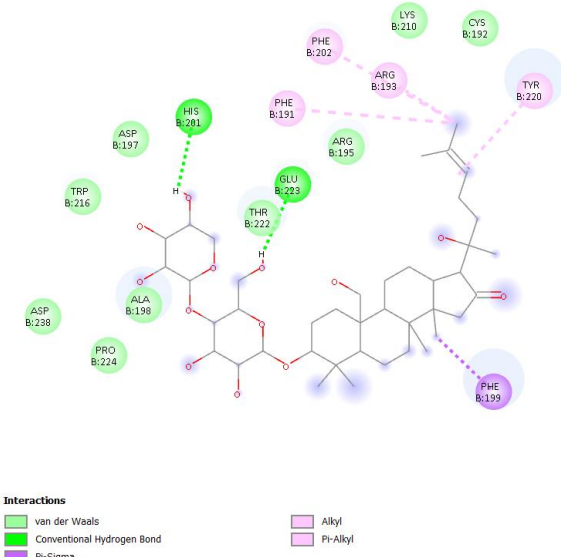


Fig : Bacoside B – 4XBM

Comparative Interpretation

Bacoside A binds slightly better to PDB ID 4XBM than Bacoside B, as shown by its docking score of -7.6 kcal/mol compared to -7.1 kcal/mol.

Both ligands form multiple hydrogen bonds and show hydrophobic as well as π -type interactions, which means both are able to fit reasonably well into the binding pocket. However, Bacoside A shows a slightly better overall interaction pattern, with well-distributed hydrogen bonds and good pocket coverage. Although it has a small unfavorable interaction, it still maintains better stability overall. Bacoside B also binds well and shows similar types of interactions, but its overall binding is slightly weaker compared to Bacoside A.

10. p21Ras (GTPase) in complex with GPPNHP (Signaling protein): [PDB ID – 1CTQ]

Bacoside A binds much better to PDB ID 1CTQ than Bacoside B. This is clearly seen from the docking scores, where Bacoside A has a much lower value (-9.5 kcal/mol) compared to -7.4 kcal/mol for Bacoside B. This large difference indicates a significantly stronger and more stable interaction.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-9.5 kcal/mol	-7.4 kcal/mol
Hydrogen bond	Multiple H-bonds (LYS117, ASN85, ASP39, SER17)	Multiple H-bonds (ASN86, LYS117, GLU13, THR30)
Hydrophobic interaction	Moderate hydrophobic interactions (PRO34 region)	Moderate hydrophobic interactions

Pi-type interaction	Limited alkyl interactions	Limited
Overall assesment	Strong and stable binding — clearly superior	Moderate binding

Table 12. Interaction Comparison at [PDB ID – 1CTQ]

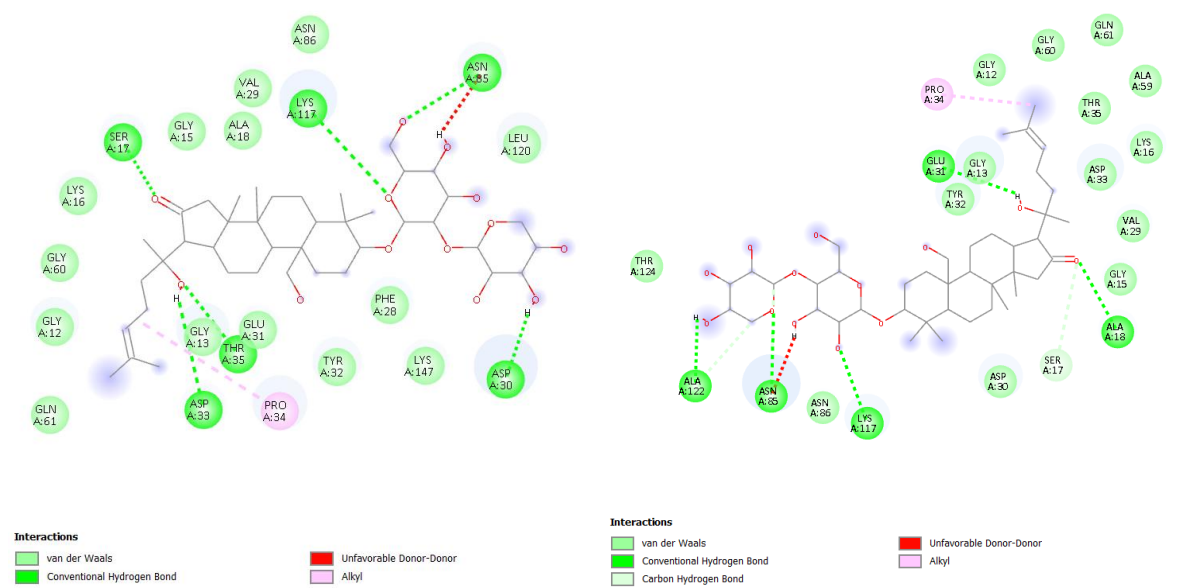


Fig : Bacoside A – 1CTQ

Fig : Bacoside B – 1CTQ

Comparative Interpretation:

Both ligands form multiple hydrogen bonds with the protein, which helps them stay in the binding site. However, Bacoside A shows a more effective and better-distributed interaction pattern, allowing it to fit more comfortably into the binding pocket.

Bacoside B also interacts with the protein through hydrogen bonds and some hydrophobic contacts, but its overall interaction network is weaker and less efficient compared to Bacoside A.

11. HES1 (Transcriptional repressor / DNA-binding protein): [PDB ID – 2MH3]

Bacoside B binds slightly better to the MH3 protein than Bacoside A. This is shown by its slightly better docking score (–8.0 kcal/mol compared to –7.9 kcal/mol for Bacoside A). The difference is small, but it still suggests a marginal improvement in binding stability.

Interaction parameter	Bacoside A	Bacoside B
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Docking score	-7.9 kcal/mol	-8.0 kcal/mol
Hydrogen bond	Multiple H-bonds (GLU43, ARG47)	Multiple H-bonds (GLU33, ARG35, GLU43)
Hydrophobic interaction	Moderate hydrophobic interactions (ALA48, HIS34 region)	Moderate hydrophobic interactions (PRO28, MET26 region)
Pi-type interaction	Present (Pi-sigma with HIS34)	Limited
Overall assesment	Strong binding	Slightly stronger binding

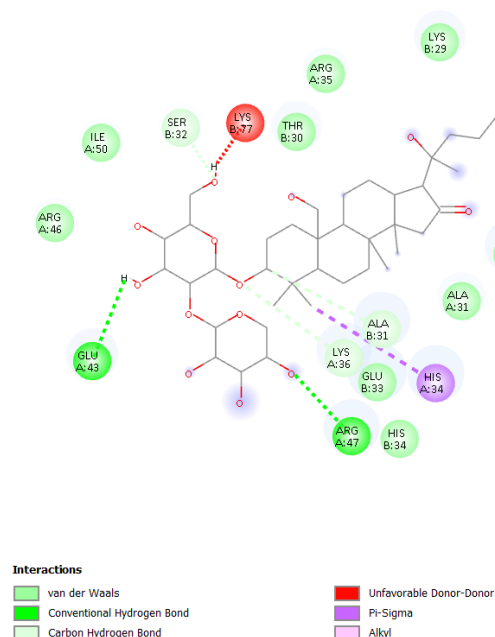


Fig : Bacoside A – 2MH3

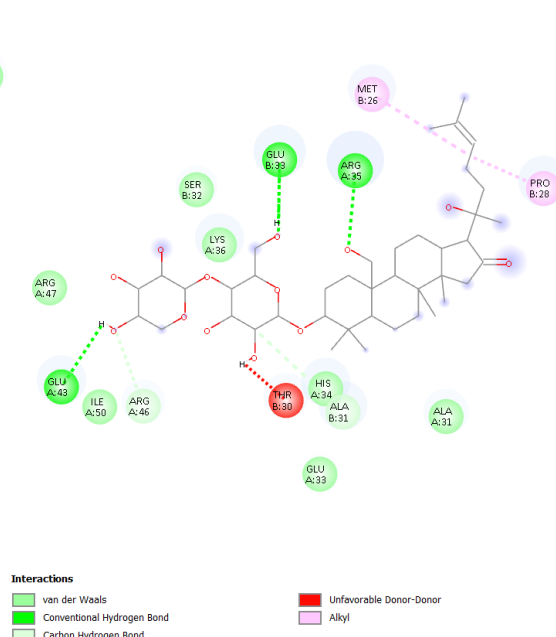


Fig : Bacoside B – 2MH3

Comparative Interpretation

Both ligands form multiple hydrogen bonds with the protein, which helps stabilize them in the binding site. However, Bacoside B forms hydrogen bonds with more residues, giving it a slightly stronger interaction network. In terms of hydrophobic interactions, both ligands show similar moderate contacts with surrounding residues. Bacoside A has an additional π -type interaction (with HIS34), which helps its stability, but this is not enough to outweigh the overall interaction pattern of Bacoside B. Both ligands also show a small unfavorable interaction, but this does not significantly affect the overall result.

12. MS0332 (Transcription-related protein): [PDB ID – 2DB7]

Docking results:

Bacoside A binds better to PDB ID 2DB7 than Bacoside B, as shown by its lower docking score (-6.5 kcal/mol compared to -5.6 kcal/mol). This indicates that Bacoside A forms a more stable interaction with the protein.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-6.5 kcal/mol	-5.6 kcal/mol
Hydrogen bond	Multiple H-bonds (GLN57, ASP62 region, water-mediated interactions)	Multiple H-bonds (TYR14, ASP13 region)
Hydrophobic interaction	Moderate hydrophobic interactions (LEU10, TYR14 region)	Moderate hydrophobic interactions (LEU17 region)
Water mediated interaction	Present (several water bridges visible)	Present but fewer
Pi type interaction	Present (alkyl / Pi interactions with aromatic residues)	Present (Pi-alkyl interactions)
Overall assesment	Moderate but better binding — superior	Weaker binding

Table 14. Interaction Comparison at MS033 [PDB ID – 2DB7]

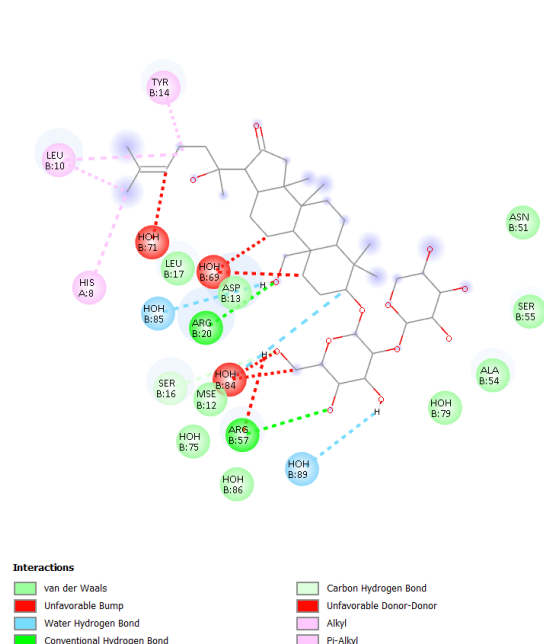


Fig : Bacoside A – 2DB7

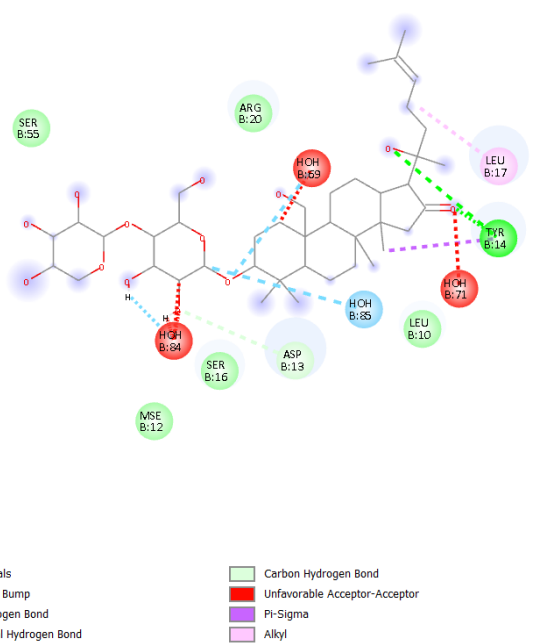


Fig : Bacoside B – 2DB7

Comparative Interpretation:

Both ligands form hydrogen bonds and hydrophobic interactions with the protein. However, Bacoside A shows a more extensive interaction network, including several water-mediated interactions, which help stabilize the ligand inside the binding site.

Bacoside B also interacts with the protein through hydrogen bonds and hydrophobic contacts, but these interactions are fewer and less well distributed. It also shows some unfavorable interactions, which may reduce its overall stability.

13. Frizzled-4 (FZD4) receptor, apo form (Wnt receptor / Membrane protein): [PDB ID – 6BD4]

Bacoside A binds better to PDB ID 6BD4 than Bacoside B. This is clear from the docking scores, where Bacoside A has a lower value (−7.7 kcal/mol) compared to −6.3 kcal/mol for Bacoside B, showing stronger and more stable binding.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-7.7 kcal/mol	−6.3 kcal/mol
Hydrogen bond	Multiple H-bonds (MET493, TRP496, ARG432)	Few H-bonds (limited interactions observed)
Hydrophobic interaction	Strong hydrophobic interactions (VAL439, LEU443, PRO447 region) ✓	Moderate hydrophobic interactions
Water mediated interaction	Not prominently observed	Not prominently observed
Pi type interaction	Present (alkyl interactions with VAL, LEU, PRO residues)	limited
Overall assesment	Strong and stable binding — superior	Weaker binding

Table 15. Interaction Comparison at (FZD4) [PDB ID – 6BD4]

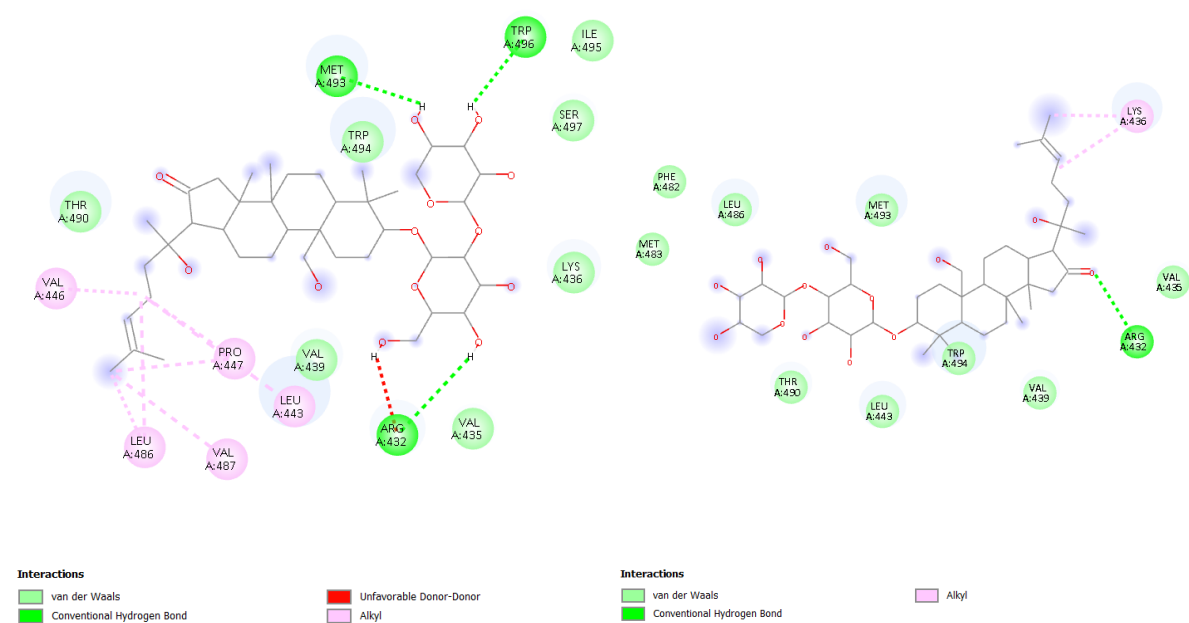


Fig : Bacoside A – 6BD4

Fig : Bacoside B – 6BD4

Comparative Interpretation

Bacoside A forms multiple hydrogen bonds with residues like MET493, TRP496, and ARG432, which help hold the ligand firmly in the binding site. It also shows strong hydrophobic interactions with residues such as VAL439, LEU443, and PRO447, allowing it to fit well into the pocket.

On the other hand, Bacoside B forms fewer hydrogen bonds and shows weaker hydrophobic interactions. Because of this, it does not interact as strongly or as effectively with the protein.

14. β -catenin (Armadillo repeat region with C-terminal domain) (Cell adhesion / Signaling protein): [PDB ID – 2Z6H]

Docking scores:

Bacoside A binds better to PDB ID 2Z6H than Bacoside B. This is supported by its lower docking score (-7.2 kcal/mol compared to -6.8 kcal/mol), which indicates stronger binding.

Interaction parameter	Bacoside A	Bacoside B
Docking score	-7.2 kcal/mol	-6.8 kcal/mol
Hydrogen bond	Multiple H-bonds (ASN287, ARG306, THR306) ✓	Multiple H-bonds (ASN426, ARG426, GLN346)

Hydrophobic interaction	Moderate hydrophobic interactions (VAL349, LYS345 region)	Moderate hydrophobic interactions
Water mediated interaction	Present (water bridges visible near binding region)	Present (water interactions observed but fewer)
Pi type interaction	Present (alkyl and pi-alkyl interactions)	Present (pi-sigma and pi-alkyl interactions)
Overall assesment	Strong and stable binding — superior	Moderate binding

Table 16. Interaction Comparison at [PDB ID – 2Z6H]

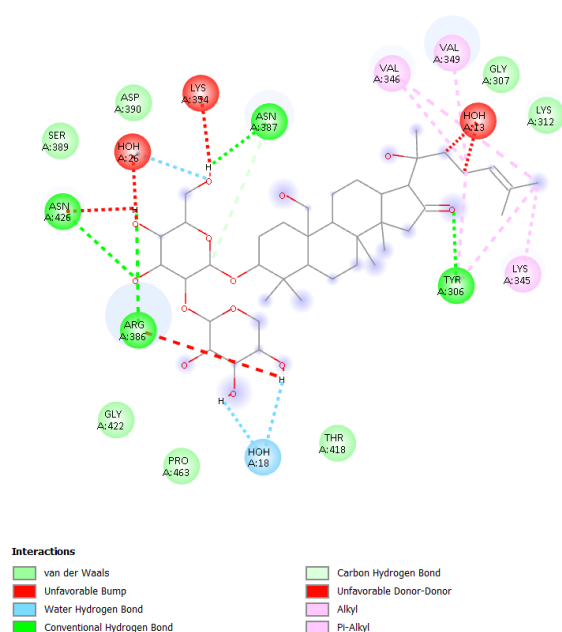


Fig : Bacoside A – 6BD4

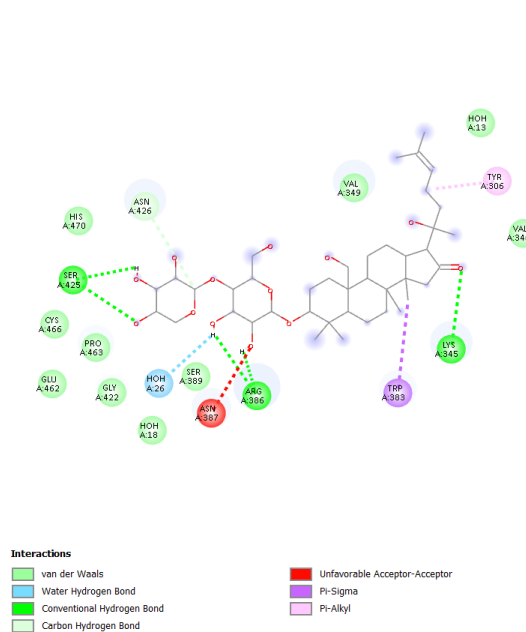


Fig : Bacoside B – 6BD4

Comparative Interpretation

Bacoside A forms several hydrogen bonds with residues like ASN287 and ARG306, helping it stay stable in the binding site. It also shows water-mediated interactions, which further support binding stability.

Bacoside B also forms hydrogen bonds and shows hydrophobic and π -type interactions, but these interactions are fewer or less effectively arranged compared to Bacoside A. It also shows an unfavorable interaction, which may reduce stability

3.4 HEPATOCELLULAR CARCINOMA (HCC):

Docking results of Liver cancer with Bacoside A and B shown that liver cancer has more stronger binding affinity scores for Bacoside B with highest score of (-11.2) with (3O96) crystal structure of human AKT1 (Protein Kinase B) is a serine/threonine kinase involved in cancer signaling . Binding affinity scores of liver cancer targets with Bacoside A had shown comparable but slightly weaker affinity scores. Bacoside B has shown consistently stronger binding (-9 to -11.2 kcal/mol) were observed with targets like (3O96, 1TUP, 1MI7,2AZ5) and better interaction across multiple liver cancer proteins.

CANCER HUB	Target Protein (PDB ID)	Bacoside B (kcal/mol)	Bacoside A (kcal/mol)	Better Binder
AKT1	3O96	-11.2	-10	Bacoside B
TP53	1TUP	-9.7	-9.6	Bacoside B
EGFR	1MI7	-9.4	-9.2	Bacoside B
TNF	2AZ5	-9	-8.3	Bacoside B
IL6	1P9M	-8.9	-8.2	Bacoside B
VEGFA	3C7Q	-8.3	-8.5	Bacoside A
HSP90AA1	3B28	-8.2	-8.4	Bacoside A
SRC	2BDJ	-7.9	-9.3	Bacoside A
CASP3	2H5I	-6.1	-6.4	Bacoside A

Table 17. Comparative Docking Binding Free Energies of Bacoside A and B(HCC)

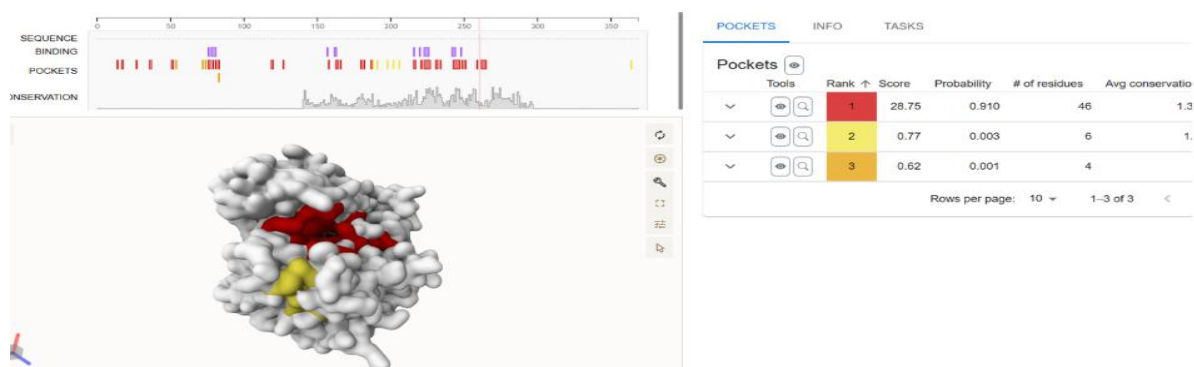


Figure : Probable pocket binding sites (3O96)using P2Rank software

These are probable pocket binding sites of 1IVO , coloured regions are various pocket binding sites each colour has different pocket locations (cavities/grooves) where ligands can bind .P2Rank analysis identified multiple potential binding pockets on the 3O96 structure, with the top-ranked pocket showing a score of 28.75 and a probability of 0.91, indicating a favorable binding region.

Binding affinity scores and Pymol visualization of Bacoside A and B with Liver cancer targets:

Protein: 1TUP

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-9.7	0.000	0.000
2	-9.4	4.400	14.495
3	-9.2	1.631	2.630
4	-8.9	5.055	9.293
5	-8.9	5.342	9.127
6	-8.9	15.508	18.462
7	-8.8	3.158	6.882
8	-8.5	15.695	21.942
9	-8.4	15.440	21.739

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-9.6	0.000	0.000
2	-9.5	1.494	2.567
3	-9.5	13.692	20.023
4	-9.3	2.257	3.432
5	-9.1	1.251	2.099
6	-8.8	1.769	2.399
7	-8.6	2.596	5.173
8	-8.6	2.718	4.337
9	-8.4	3.124	6.336

Writing output ... done.

Fig. Binding Affinity to Bacoside B

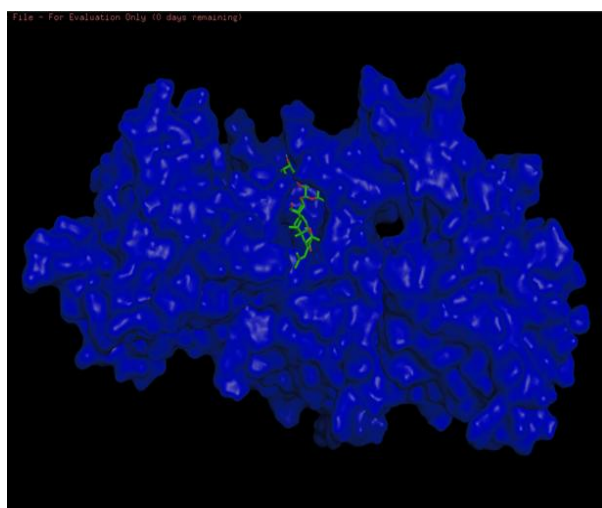


Fig. Docking site of Bacoside A

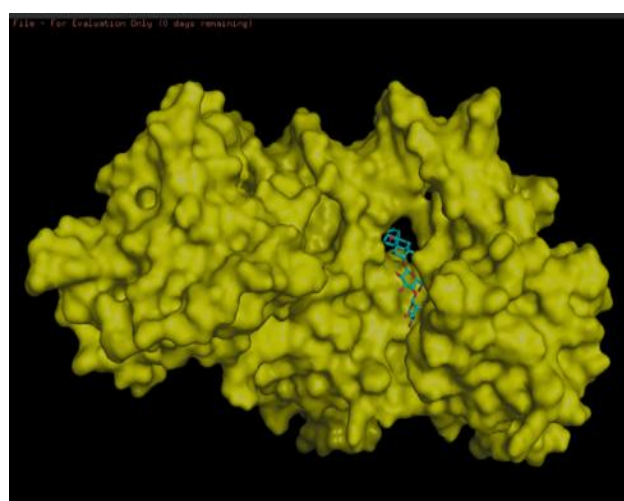


Fig. Docking site of Bacoside B

PROTEIN: 1P9M

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.2	0.000	0.000
2	-8.0	2.901	12.797
3	-7.9	4.557	13.032
4	-7.8	3.179	12.397
5	-7.6	4.251	11.953
6	-7.5	8.161	13.523
7	-7.5	5.199	12.278
8	-7.4	4.432	12.027
9	-7.4	25.963	29.561

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.9	0.000	0.000
2	-7.8	1.415	2.439
3	-7.4	2.718	6.320
4	-7.3	5.891	16.337
5	-7.3	5.385	15.970
6	-7.2	6.078	15.763
7	-7.0	3.046	5.901
8	-6.9	3.198	4.967
9	-6.8	2.550	5.632

Writing output ... done.

Fig. Binding Affinity to Bacoside B

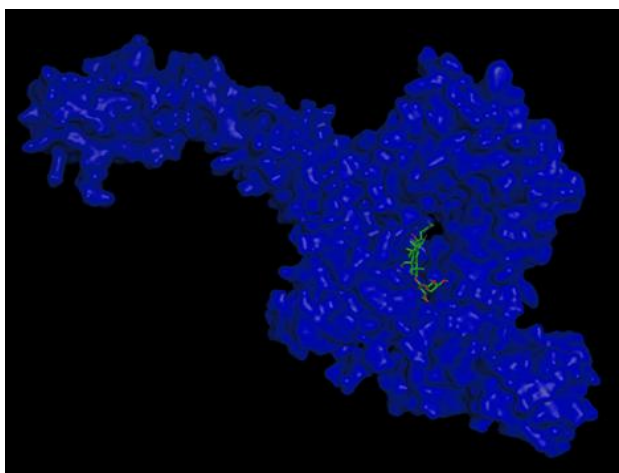


Fig. Docking site of Bacoside A

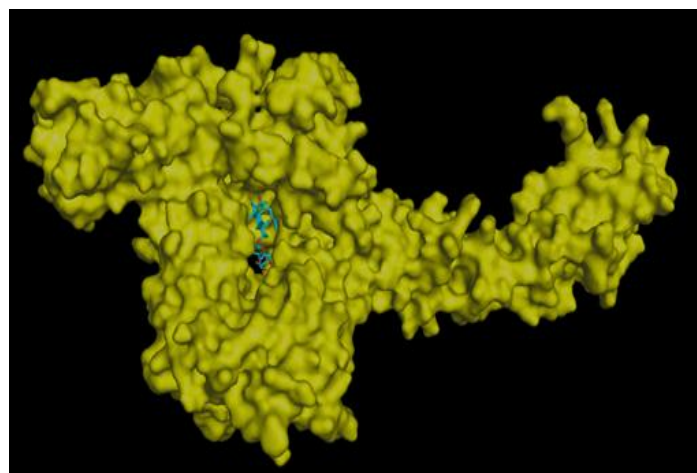


Fig. Docking site of Bacoside B

PROTEIN: 3O96

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-10.0	0.000	0.000
2	-9.6	14.938	19.408
3	-9.3	6.641	11.515
4	-9.2	9.199	15.406
5	-9.1	4.991	12.754
6	-8.9	15.847	19.319
7	-8.9	15.527	19.904
8	-8.9	8.269	14.005
9	-8.8	15.405	19.937

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-11.2	0.000	0.000
2	-8.3	16.879	22.416
3	-8.0	2.739	6.197
4	-7.8	13.894	16.803
5	-7.7	15.776	18.249
6	-7.7	17.347	23.082
7	-7.6	11.784	15.755
8	-7.6	14.304	20.069
9	-7.6	16.140	22.556

Writing output ... done.

Fig. Binding Affinity to Bacoside B

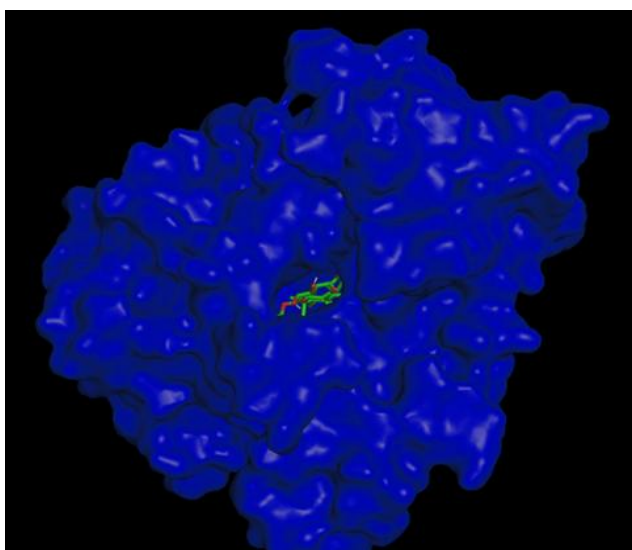


Fig. Docking site of Bacoside A

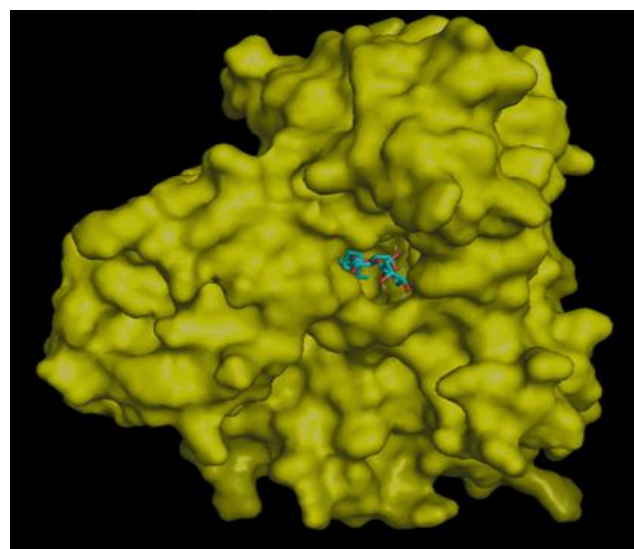


Fig. Docking site of Bacoside B

PROTEIN: 2H5I

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-6.4	0.000	0.000
2	-6.1	7.997	12.837
3	-6.1	27.073	30.463
4	-6.0	7.759	11.896
5	-5.9	2.548	4.011
6	-5.9	5.116	11.703
7	-5.9	7.305	12.334
8	-5.8	9.353	13.595
9	-5.8	3.488	12.526

Writing output ... done.

Fig. Binding Affinity to Bacoside A

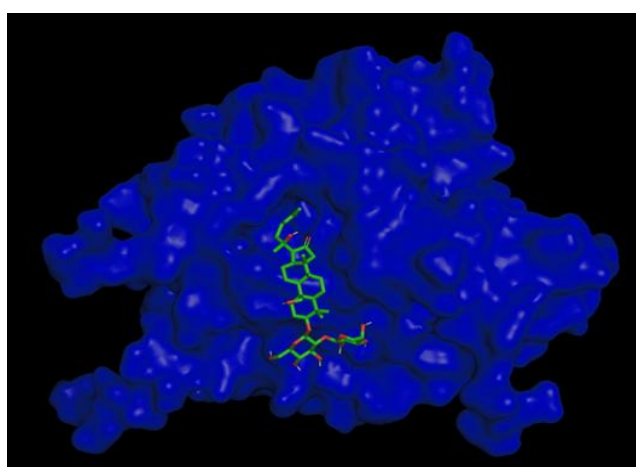


Fig. Docking site of Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-6.1	0.000	0.000
2	-5.9	5.326	12.381
3	-5.9	32.253	34.802
4	-5.7	31.944	34.592
5	-5.6	1.637	2.474
6	-5.5	30.977	34.749
7	-5.4	9.677	12.374
8	-5.2	8.303	18.374
9	-5.2	8.365	11.138

Writing output ... done.

Fig. Binding Affinity to Bacoside B

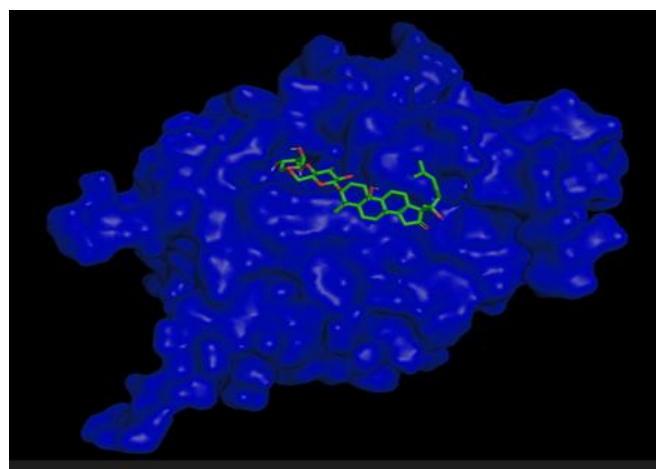


Fig. Docking site of Bacoside B

PROTEIN: 2AZ5

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.3	0.000	0.000
2	-8.3	2.347	3.304
3	-8.0	2.513	10.090
4	-8.0	3.137	10.288
5	-7.9	2.407	9.910
6	-7.9	2.790	4.832
7	-7.9	2.007	4.481
8	-7.8	2.554	10.577
9	-7.8	2.580	4.808

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-9.0	0.000	0.000
2	-8.8	18.205	22.913
3	-8.3	1.839	2.727
4	-8.2	18.664	25.416
5	-8.2	2.164	3.442
6	-8.2	17.860	22.481
7	-8.1	4.370	6.377
8	-8.0	3.619	5.582
9	-7.9	18.501	25.142

Writing output ... done.

Fig. Binding Affinity to Bacoside B

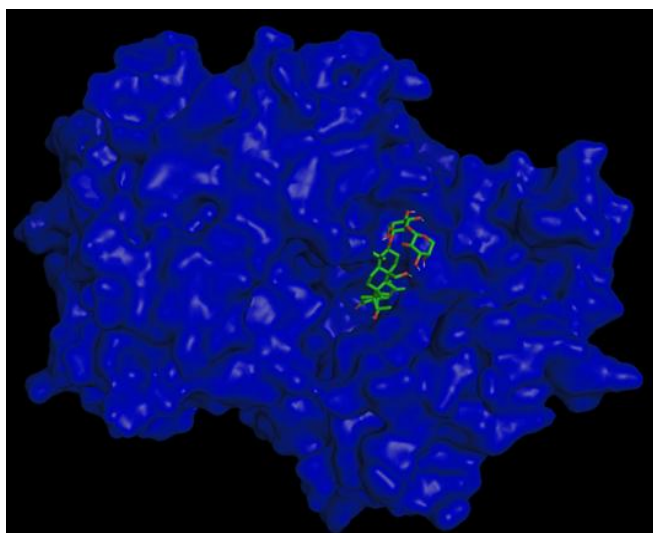


Fig. Docking site of Bacoside B

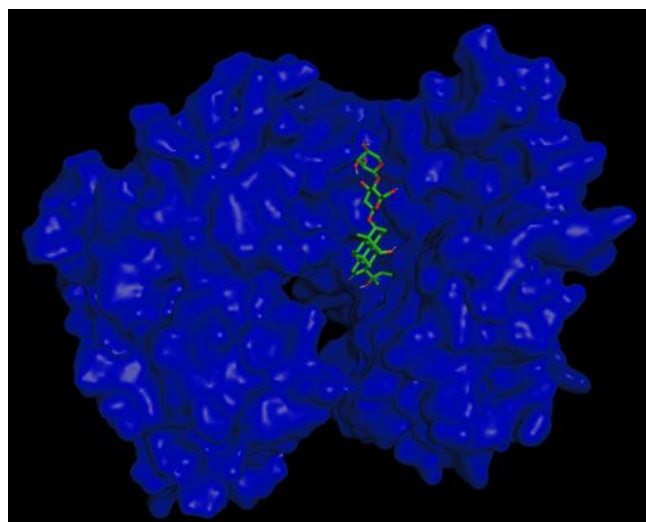


Fig. Docking site of Bacoside A

PROTEIN: 2BDJ

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-9.3	0.000	0.000
2	-9.1	5.353	9.474
3	-8.9	8.647	12.501
4	-8.7	2.659	4.557
5	-8.2	5.098	8.211
6	-7.9	2.393	3.764
7	-7.9	1.658	2.247
8	-7.8	5.698	9.486
9	-7.8	6.968	10.099

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-7.9	0.000	0.000
2	-7.6	10.884	20.372
3	-7.4	9.652	15.534
4	-7.4	5.290	8.751
5	-7.0	9.138	20.012
6	-7.0	5.436	12.591
7	-7.0	5.134	8.885
8	-6.8	1.938	2.777
9	-6.8	4.861	9.141

Writing output ... done.

Fig. Binding Affinity to Bacoside B

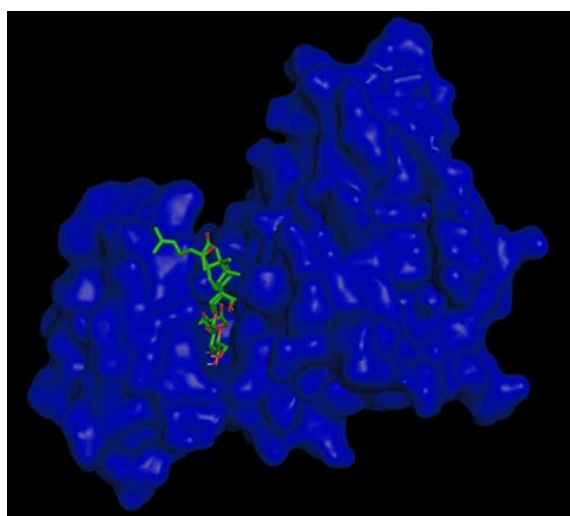


Fig. Docking site of Bacoside A

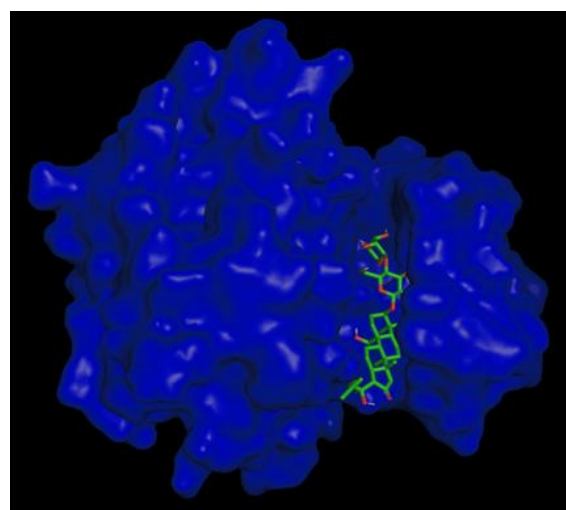


Fig. Docking site of Bacoside B

PROTEIN: 3B28

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.4	0.000	0.000
2	-8.2	3.221	4.736
3	-8.1	2.677	3.615
4	-8.1	3.193	5.384
5	-8.1	1.167	1.568
6	-8.0	4.287	7.266
7	-7.9	2.180	2.937
8	-7.8	2.237	4.794
9	-7.7	14.285	18.302

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.2	0.000	0.000
2	-7.9	4.641	8.382
3	-7.9	5.127	9.032
4	-7.9	4.815	8.390
5	-7.8	16.367	18.969
6	-7.7	1.586	2.179
7	-7.5	17.470	20.146
8	-7.4	6.858	10.390
9	-7.3	2.393	3.817

Writing output ... done.

Fig. Binding Affinity to Bacoside B

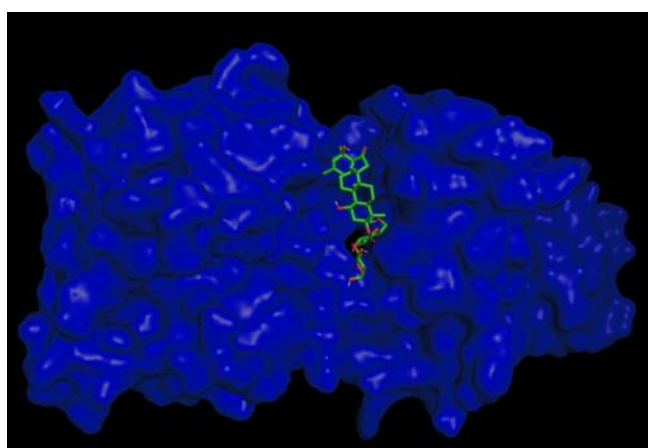


Fig. Docking site of Bacoside A

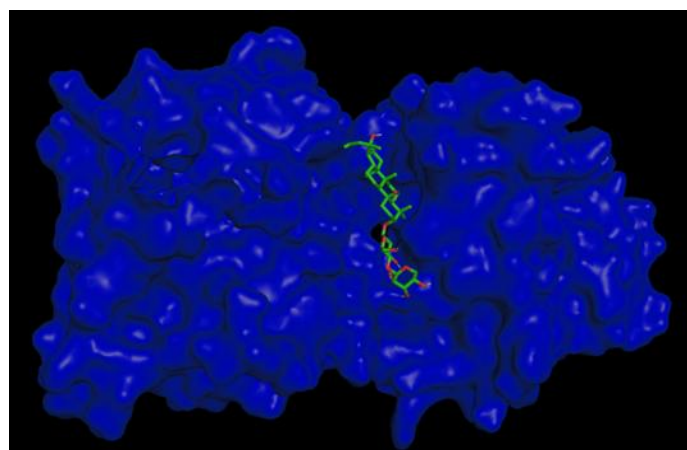


Fig. Docking site of Bacoside B

PROTEIN: 3C7Q

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.5	0.000	0.000
2	-8.4	3.787	7.312
3	-8.4	2.814	10.372
4	-8.3	7.631	11.319
5	-8.3	8.550	14.478
6	-8.3	7.663	13.315
7	-8.1	10.980	15.362
8	-8.1	6.125	12.110
9	-8.1	7.426	12.210

Writing output ... done.

Fig. Binding Affinity to Bacoside A

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-8.3	0.000	0.000
2	-7.6	1.611	2.521
3	-7.5	4.163	6.727
4	-7.4	13.915	17.643
5	-7.4	1.985	3.221
6	-7.1	1.836	3.403
7	-7.0	3.283	5.620
8	-6.9	4.128	12.884
9	-6.8	4.209	6.335

Writing output ... done.

Fig. Binding Affinity to Bacoside B

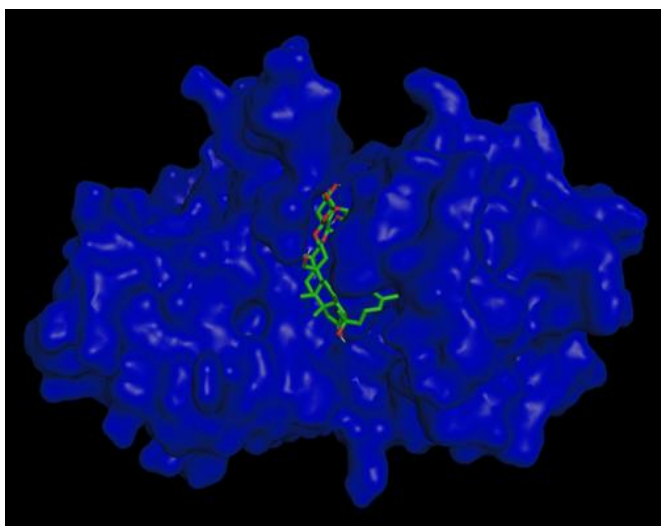


Fig. Docking site of Bacoside A

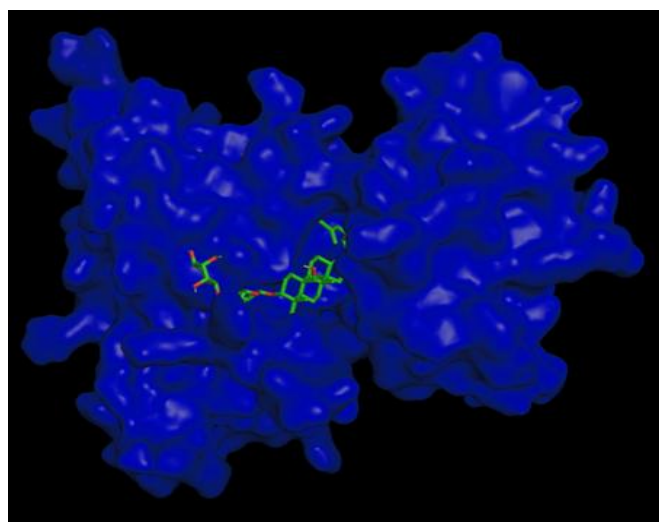


Fig. Docking site of Bacoside B

PROTEIN: 1M17

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-9.2	0.000	0.000
2	-9.0	1.362	3.806
3	-9.0	1.047	1.087
4	-8.9	1.844	4.447
5	-8.9	3.148	12.054
6	-8.9	1.808	4.447
7	-8.8	1.689	2.498
8	-8.8	1.967	2.847
9	-8.8	2.301	4.599

Writing output ... done.

mode	affinity (kcal/mol)	dist from best mode	
		rmsd l.b.	rmsd u.b.
1	-9.4	0.000	0.000
2	-9.1	1.334	2.085
3	-8.9	1.719	2.201
4	-8.7	7.531	11.222
5	-8.7	6.983	9.987
6	-8.6	2.486	10.900
7	-8.3	7.827	12.851
8	-8.2	2.962	11.641
9	-8.1	2.368	2.967

Writing output ... done.

Fig. Binding Affinity to Bacoside A

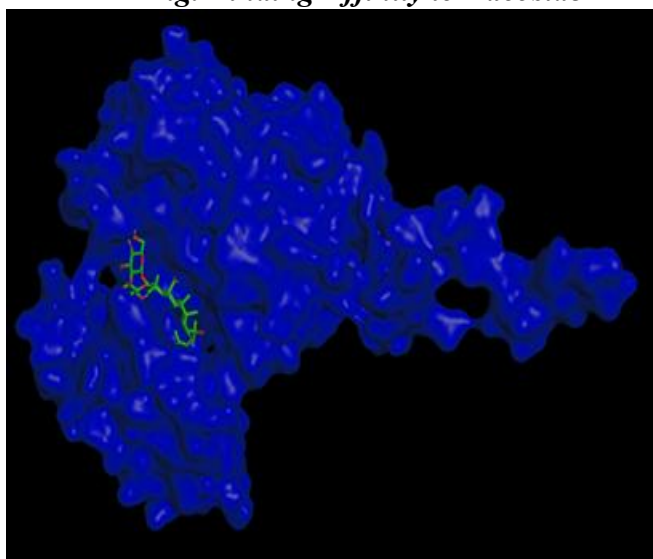


Fig. Binding Affinity to Bacoside B

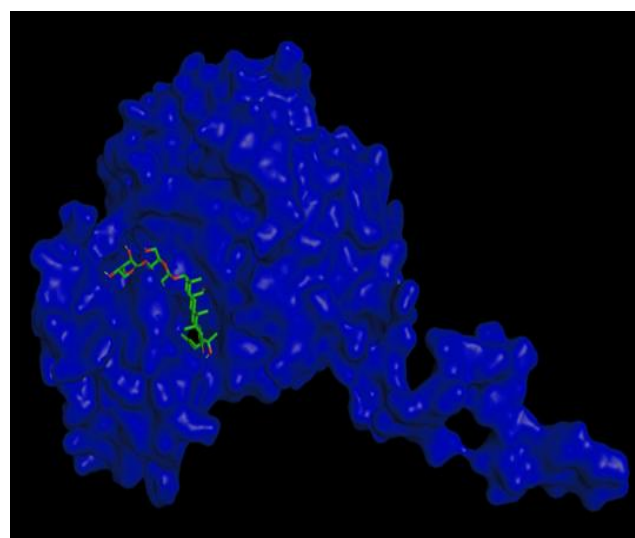


Fig. Docking site of Bacoside A

Fig. Docking site of Bacoside B

3.5 Protein Ligand interaction analysis using Discovery Studio

This section presents a detailed analysis of the docking results and 2D interaction maps for each of the nine cancer hub proteins. For each protein, the biological context is established, followed by individual analysis of Bacoside A and B binding, comparative interpretation, and a summary interaction table. The 2D interaction maps from BIOVIA Discovery Studio are presented alongside the analysis.

Hepatocellular Carcinoma:

1. AKT1 / Protein Kinase B (PDB: 3O96)

Biological Background

AKT1 (also known as Protein Kinase B) is a serine/threonine kinase that plays a central role in the PI3K/AKT/mTOR signaling axis it is one of the most frequently dysregulated pathways in human cancer. AKT1 promotes cell survival by phosphorylating and inactivating pro-apoptotic proteins such as BAD and FOXO transcription factors, while simultaneously activating downstream effectors including mTORC1, which drives protein synthesis and cellular growth. AKT1 is overexpressed or hyperactivated in breast, ovarian, pancreatic, and prostate cancers. Its ATP-binding kinase domain is the primary pharmacological target for inhibitor development.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	-10.0 kcal/mol	-11.2 kcal/mol ✓
Hydrogen Bonds	Moderate number with active site residues	Higher number, including key gatekeeper contacts
Hydrophobic Contacts	Good coverage of hydrophobic pocket	Extensive, more residues engaged
Pi-type Interactions	Pi-alkyl contacts present	Additional pi-alkyl and pi-sigma interactions
Overall Assessment	Strong binding	Superior binding — best in entire dataset

Table 18. Interaction Comparison at AKT1

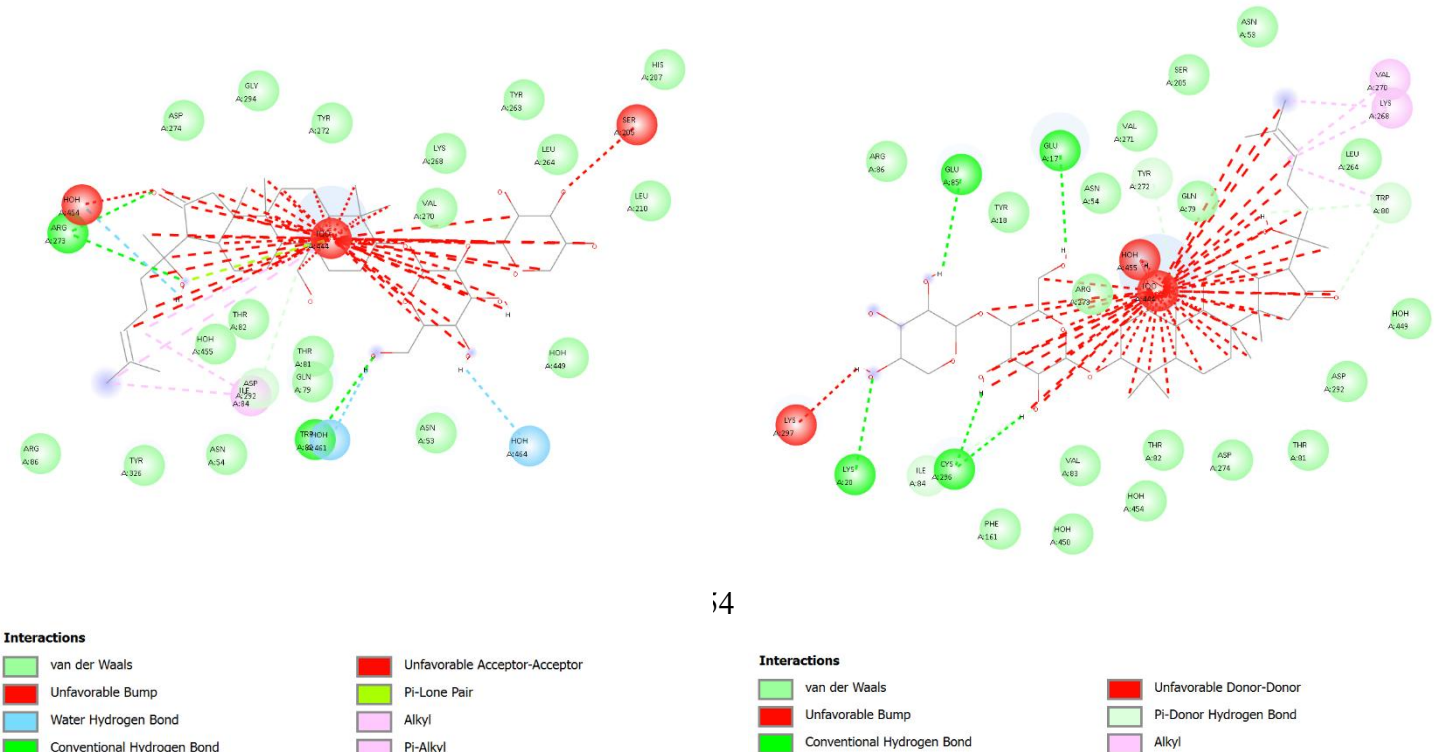


Fig: 1A. Bacoside A – AKT1 (3O96)**Fig :1B. Bacoside B – AKT1 (3O96)****Comparative Interpretation:**

The 1.2 kcal/mol advantage of Bacoside B over Bacoside A at AKT1 is corroborated by the Discovery Studio 2D maps. Bacoside B establishes a denser network of contacts within the ATP-binding kinase domain, particularly through enhanced hydrophobic contacts with residues lining the deep hydrophobic back pocket, a pharmacophoric feature critical for potent kinase inhibition. The additional pi-alkyl interactions visible in Bacoside B's map reflect its slightly different stereochemical presentation, allowing it to fill the kinase pocket more completely. Given AKT1's central role in cancer cell survival, Bacoside B's superior binding here has significant pharmacological implications for anti-proliferative and pro-apoptotic activity.

2. TP53 / Tumor Protein p53 (PDB: 1TUP)**Biological Background**

TP53 is arguably the most important tumor suppressor protein in the human genome, earning the designation "guardian of the genome." It is a transcription factor that responds to cellular stress — DNA damage, hypoxia, oncogene activation — by activating target genes that induce cell cycle arrest, DNA repair, senescence, or apoptosis. TP53 is mutated in more than 50% of all human cancers, making it the most frequently altered gene in oncology. Restoration of TP53 function or targeting TP53 in its wild-type form represents a major avenue of cancer drug discovery. The 1TUP crystal structure captures the TP53 DNA-binding domain in complex with DNA and zinc.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	–9.6 kcal/mol	–9.7 kcal/mol (marginally better)
Hydrogen Bonds	Multiple H-bonds with DNA-binding residues	Comparable H-bond network
Hydrophobic Contacts	Good hydrophobic pocket coverage	Similar coverage
Zinc Coordination Region	Contacts near zinc-coordinating residues	Similar zinc region contacts
Overall Assessment	Essentially equivalent — within error margin	Essentially equivalent — within error margin

Table 19. Interaction Comparison at 1TUP

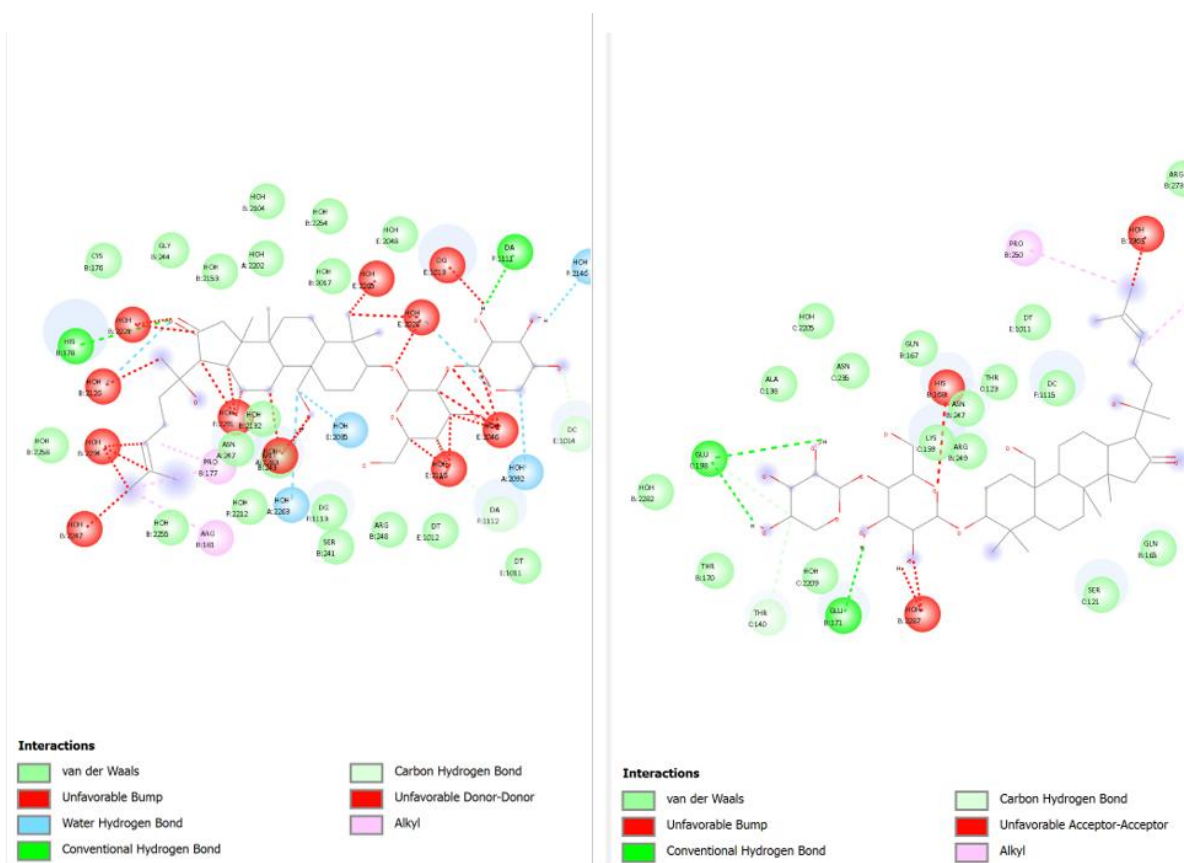


Fig:2A. Bacoside A – TP53 (PDB: 1TUP) Fig:2B Bacoside B – TP53 (PDB: 1TUP)

Comparative Interpretation

The 0.1 kcal/mol difference between the two compounds at TP53 falls well within the accepted error margin of standard docking algorithms (typically ± 0.5 kcal/mol or greater). Both compounds should therefore be considered to bind TP53 with equivalent affinity. The 2D interaction maps confirm this equivalence, showing comparable hydrogen bond patterns with the DNA-binding domain residues and similar hydrophobic contact distributions. From a pharmacological perspective, both Bacoside A and B may be considered equally promising candidates for TP53 modulation, which is particularly significant given the near-universal importance of TP53 in cancer biology.

3. EGFR / Epidermal Growth Factor Receptor (PDB: 1M17)

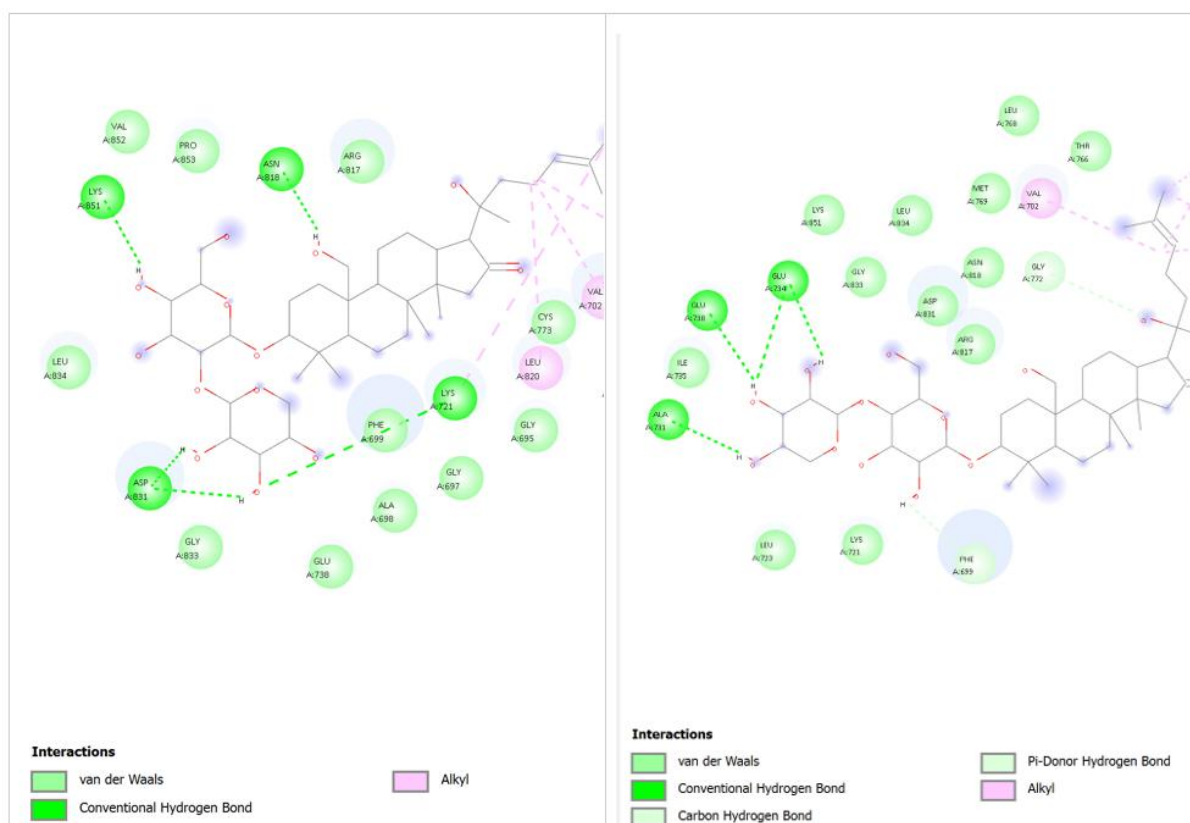
Biological Background

EGFR (Epidermal Growth Factor Receptor) is a receptor tyrosine kinase belonging to the ErbB family. Upon ligand binding, EGFR dimerizes and autophosphorylates, triggering downstream signaling cascades including the RAS/MAPK, PI3K/AKT, and STAT pathways that promote cell proliferation, survival, and differentiation. EGFR is overexpressed or mutated in non-small cell lung cancer (NSCLC), colorectal cancer, head and neck squamous cell carcinoma, and glioblastoma. Targeted EGFR inhibitors (erlotinib, gefitinib, osimertinib) have transformed treatment of EGFR-mutant NSCLC. The 1M17 structure represents the EGFR kinase domain in its active conformation

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	−9.2 kcal/mol	−9.4 kcal/mol ✓
Hinge Region H-bonds	Present with hinge backbone residues	Comparable, potentially additional contact
Hydrophobic Back Pocket	Good engagement	Slightly better complementarity
Gatekeeper Residue	Contacts present	Contacts present
Overall Assessment	Strong EGFR binding	Marginally stronger EGFR binding

Table 20. Interaction Comparison at 1M17



Comparative Interpretation

Both compounds engage the EGFR ATP-binding pocket in a manner consistent with clinically relevant kinase inhibitors, forming hydrogen bonds with hinge backbone residues (critical for ATP-competitive inhibition) and establishing hydrophobic contacts with the deep hydrophobic pocket. Bacoside B's marginal advantage at −9.4 kcal/mol may reflect slightly better shape complementarity with the EGFR binding site geometry. The comparable binding energies of both compounds at EGFR suggest that either could serve as a molecular scaffold for developing natural product-derived EGFR-targeted agents, particularly relevant in the context of NSCLC treatment where EGFR inhibitors are standard of care.

4. TNF / Tumor Necrosis Factor Alpha (PDB: 2AZ5)

Biological Background

Tumor Necrosis Factor Alpha (TNF- α) is a pleiotropic cytokine produced primarily by macrophages, monocytes, and T lymphocytes. While originally characterized for its anti-tumor properties (capable of inducing tumor cell necrosis at high concentrations), sustained low-level TNF- α signaling in the tumor microenvironment (TME) promotes tumor progression through NF- κ B activation, angiogenesis, immunosuppression, and induction of epithelial-mesenchymal transition. Elevated TNF- α levels correlate with poor prognosis in multiple cancers including breast, colorectal, and pancreatic cancers. The 2AZ5 structure captures the TNF trimer in its bioactive homotrimeric form.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	-8.3 kcal/mol	-9.0 kcal/mol ✓
Hydrogen Bonds	H-bonds with TNF interface residues	More extensive H-bond network at trimer interface
Hydrophobic Contacts	Moderate contacts	Denser hydrophobic contacts within trimer pocket
Trimer Interface Penetration	Partial penetration	Deeper penetration into trimeric interface
Overall Assessment	Moderate TNF binding	Strong TNF binding — 0.7 kcal/mol advantage

Table 21. Interaction Comparison at 2AZ5

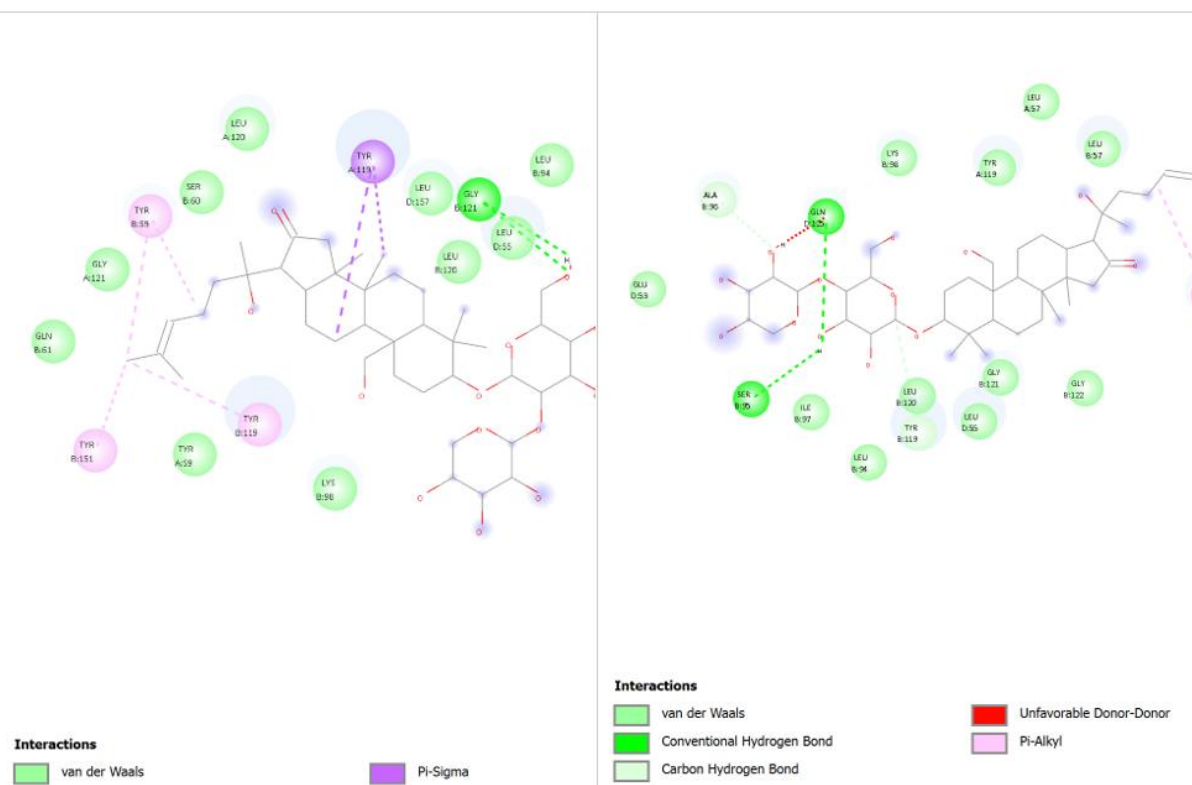


Fig:4A. Bacoside A-TNF(PDB: 2AZ5)

Fig:4B. Bacoside B-TNF(PDB: 2AZ5)

Comparative Interpretation

The TNF homotrimer presents a unique binding pocket at the interface between adjacent monomers, distinct from typical enzyme active sites. Bacoside B's 0.7 kcal/mol advantage suggests it achieves deeper penetration into this trimeric interface pocket, establishing more hydrogen bonds with interfacial residues and denser hydrophobic contacts within the relatively enclosed cavity. Interfacial pocket inhibitors of TNF represent an important but challenging drug class, natural products with the ability to engage this pocket are of high pharmacological interest. Bacoside B's superior binding here, combined with its similar advantage at IL6, suggests a particular affinity for cytokine-type binding interfaces and a potential anti-inflammatory role within the tumor microenvironment.

5. IL6 / Interleukin-6 (PDB: 1P9M)

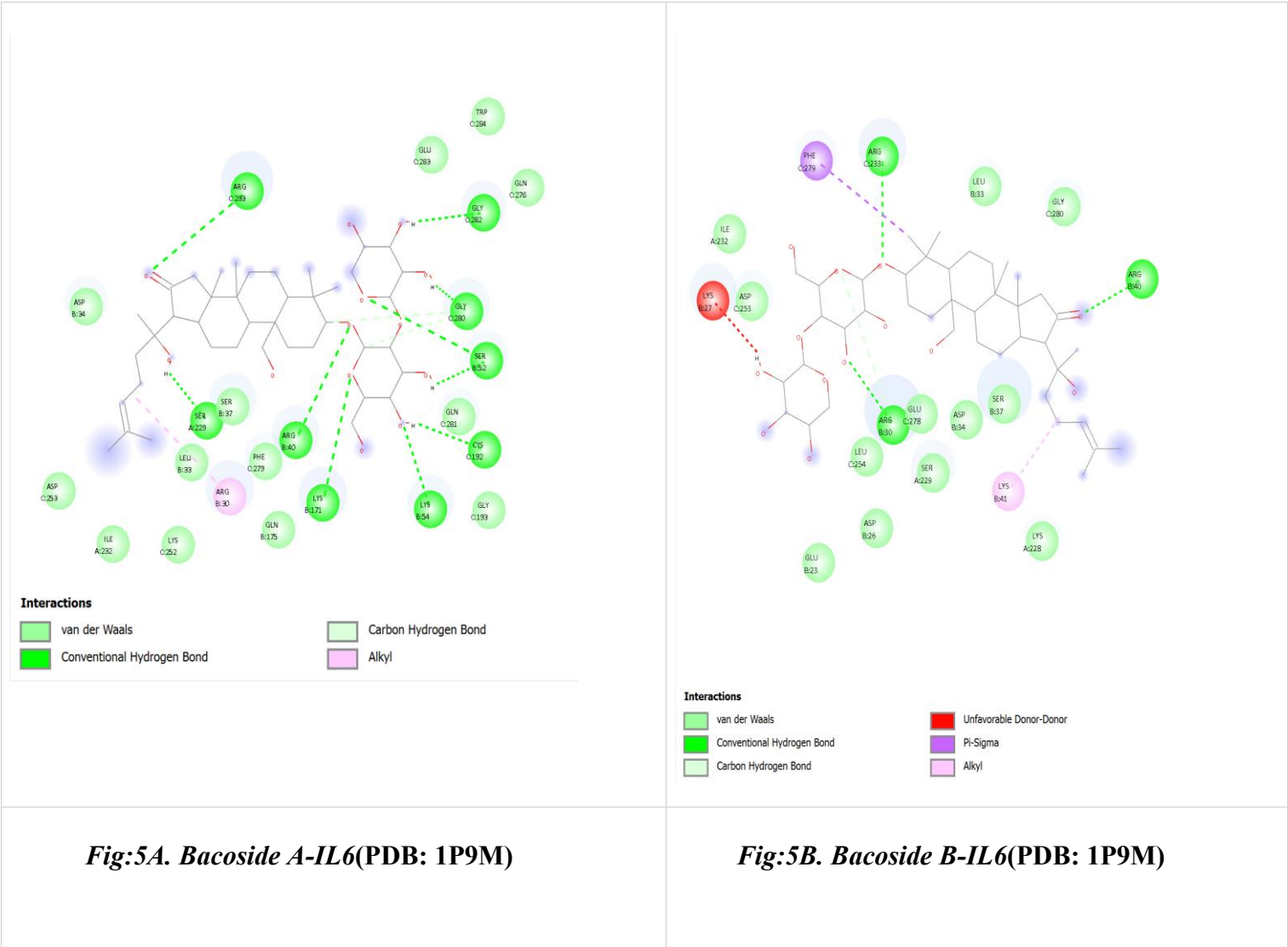
Biological Background

Interleukin-6 (IL-6) is a multifunctional cytokine with both pro-inflammatory and anti-inflammatory properties, produced by fibroblasts, macrophages, lymphocytes, and tumor cells. In cancer, IL-6 plays a particularly important role in activating the JAK/STAT3 signaling axis, promoting tumor cell proliferation, survival, angiogenesis, and immune evasion. Elevated circulating IL-6 is associated with poor prognosis in multiple myeloma, colorectal, renal cell, ovarian, and lung cancers. The tumor microenvironment often features autocrine and paracrine IL-6 signaling that creates an immunosuppressive niche. Therapeutic anti-IL6 antibodies (tocilizumab, siltuximab) are clinically validated in oncology.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	−9.2 kcal/mol	−9.4 kcal/mol ✓
Hinge Region H-bonds	Present with hinge backbone residues	Comparable, potentially additional contact
Hydrophobic Back Pocket	Good engagement	Slightly better complementarity
Gatekeeper Residue	Contacts present	Contacts present
Overall Assessment	Strong EGFR binding	Marginally stronger EGFR binding

Table 22. Interaction Comparison at 1P9M



Interactions

- van der Waals
- Conventional Hydrogen Bond
- Carbon Hydrogen Bond
- Unfavorable Donor-Donor
- Pi-Sigma
- Alkyl

Comparative Interpretation

The consistent 0.7 kcal/mol advantage of Bacoside B at both IL6 and TNF is noteworthy because it suggests a structurally-based preference of Bacoside B for cytokine binding interfaces over kinase catalytic sites. Both cytokines share protein-protein interaction (PPI) type binding sites over shallow, large-surface-area pockets rather than deep enzymatic clefts,

which may be particularly well-complemented by the slightly larger or differently oriented sugar moieties of Bacoside B. Simultaneously targeting both TNF and IL6 would offer broad-spectrum suppression of pro-tumorigenic cytokine signaling, and Bacoside B's superior binding at both makes it a candidate for tumor microenvironment modulation.

6. VEGFA / Vascular Endothelial Growth Factor A (PDB: 3C7Q)

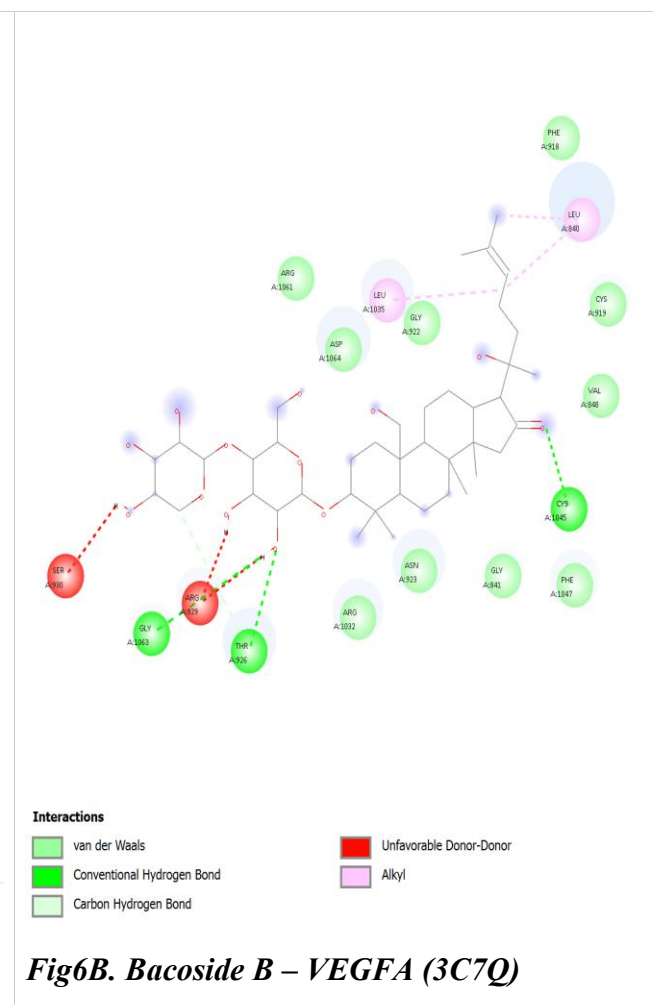
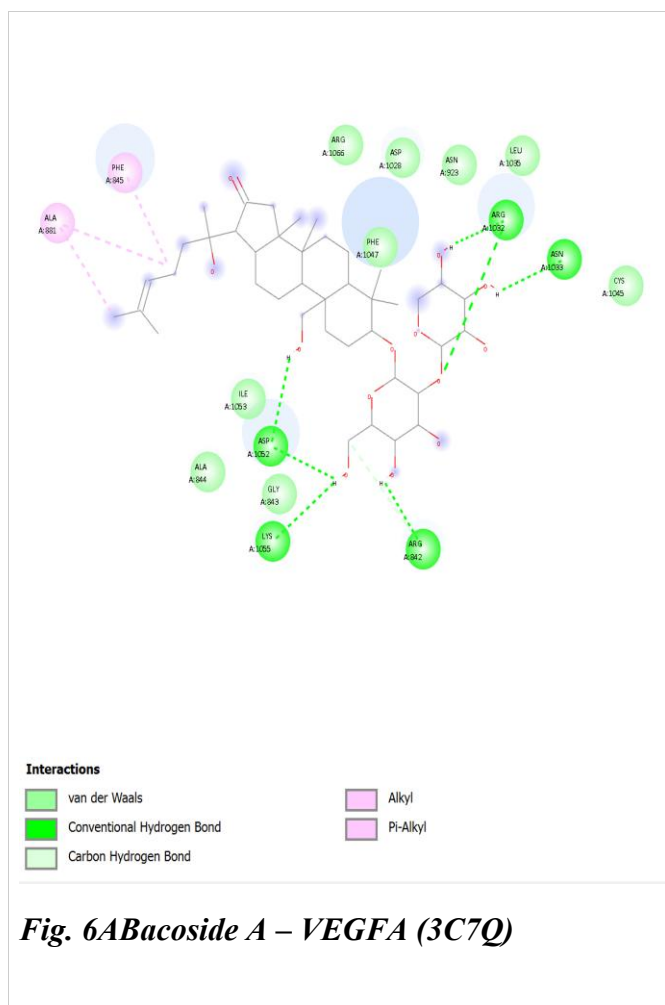
Biological Background

VEGFA (Vascular Endothelial Growth Factor A) is the primary driver of tumor angiogenesis, the process by which tumors develop their own blood supply. Without new blood vessel formation, tumors cannot grow beyond a few millimeters. VEGFA binds to VEGFR1 and VEGFR2 on endothelial cells, triggering signaling cascades that promote endothelial cell proliferation, migration, and tube formation. VEGFA is overexpressed in virtually all solid tumors and correlates with tumor stage, metastatic potential, and poor prognosis. Anti-VEGFA therapy (bevacizumab) is an established clinical strategy in colorectal, lung, and renal cancers. The 3C7Q structure captures VEGFA in its dimeric form.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	-8.5 kcal/mol ✓	-8.3 kcal/mol
Hydrogen Bonds	H-bonds with receptor-binding loop residues	Fewer H-bonds with binding interface
Hydrophobic Contacts	Good contact with hydrophobic interface patches	Comparable but slightly fewer contacts
Receptor Interface Engagement	Better engagement of VEGFR-binding epitopes	Partial engagement
Overall Assessment	Better anti-angiogenic potential	Moderate VEGFA binding

Table 23. Interaction Comparison at 3C7Q

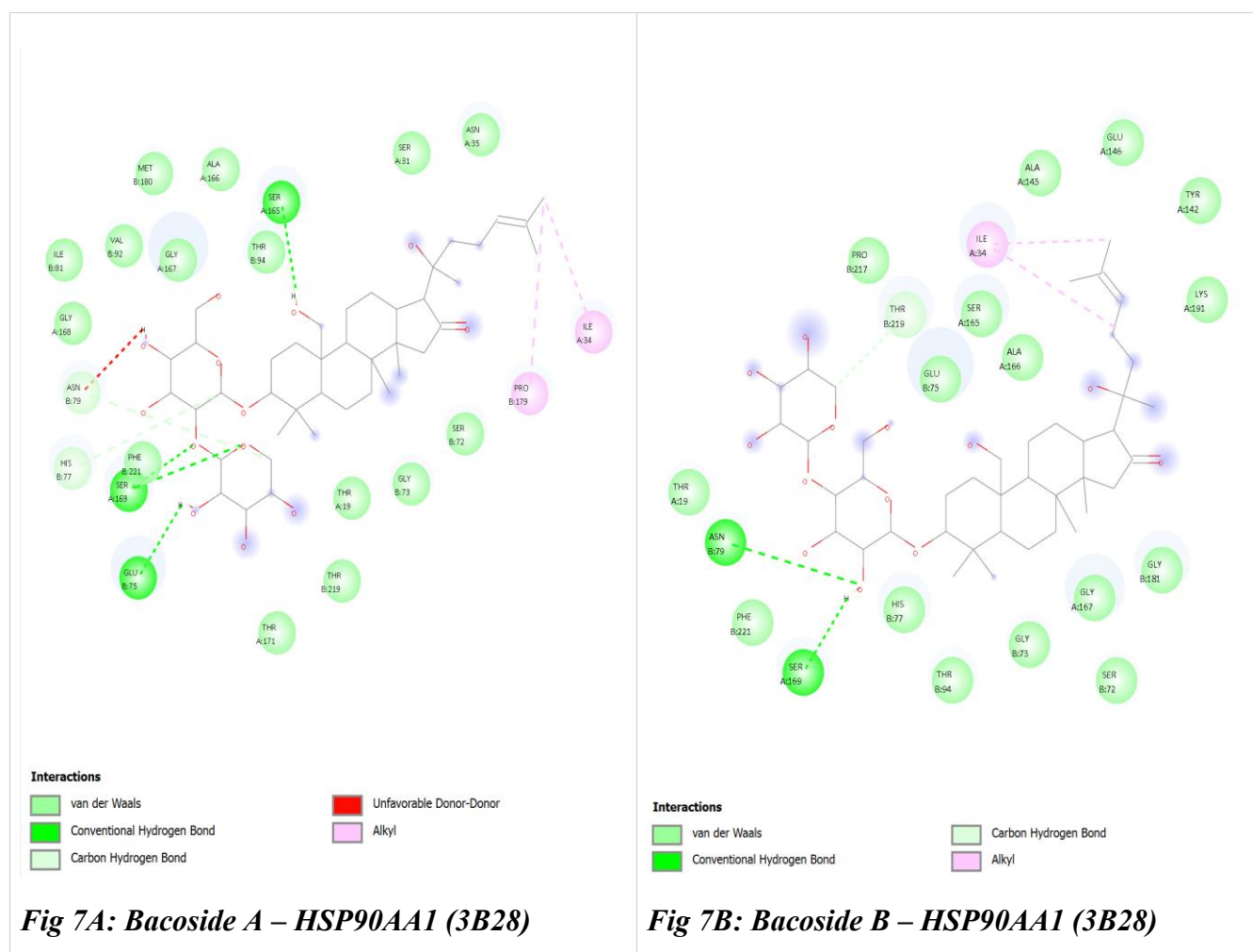


3B28) is the primary druggable site for HSP90 inhibitors such as geldanamycin and its clinical derivatives.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	−8.4 kcal/mol ✓	−8.2 kcal/mol
Hydrogen Bonds	H-bonds with N-terminal ATP pocket residues (Asp93, Gly97)	Comparable H-bonds, slightly shifted pose
Hydrophobic Contacts	Good engagement of hydrophobic shelf	Similar but slightly fewer contacts
ATP Pocket Engagement	Better fit within the adenine-binding pocket	Partially shifted binding pose
Overall Assessment	Better HSP90 inhibitory potential	Good but marginally weaker binding

Table 24. Interaction Comparison at 3B28



Comparative Interpretation

The N-terminal ATP pocket of HSP90 features a conserved water-mediated hydrogen bond network and a distinctive "lid" region. Bacoside A's slight advantage at this target suggests its

trisaccharide chain interacts more favorably with the specific geometry of the HSP90 ATP-binding cleft. The implications are noteworthy: HSP90 inhibition leads to simultaneous destabilization and degradation of multiple oncoproteins (HER2, AKT, CDK4), effectively amplifying the anticancer effect through client protein degradation. Bacoside A's stronger HSP90 binding thus predicts a broad, multi-oncoprotein suppression effect that extends beyond simple single-target inhibition.

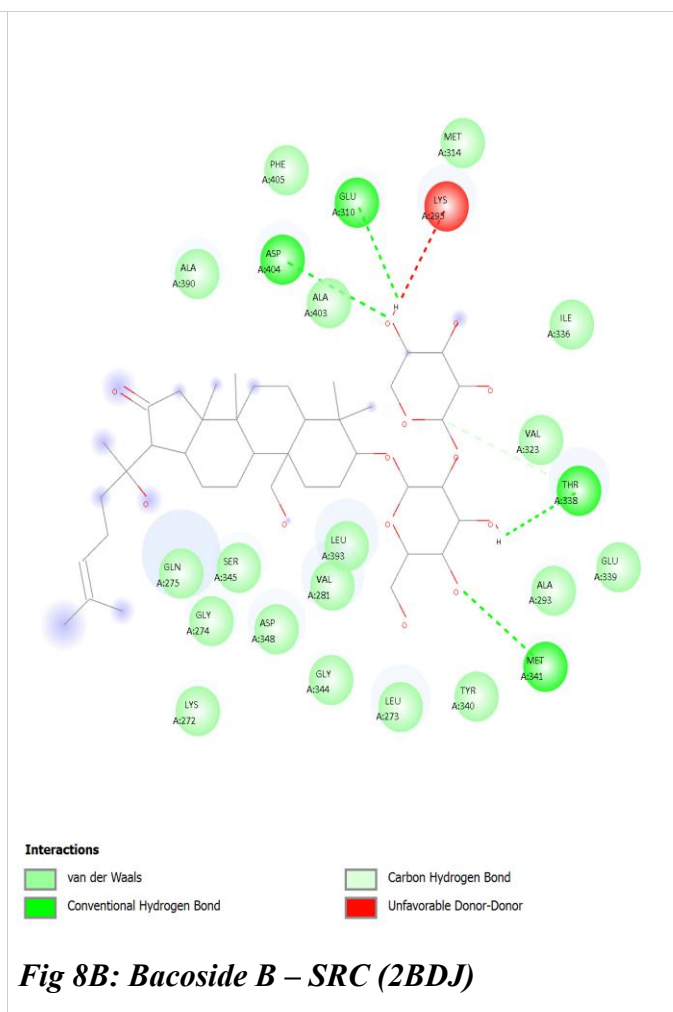
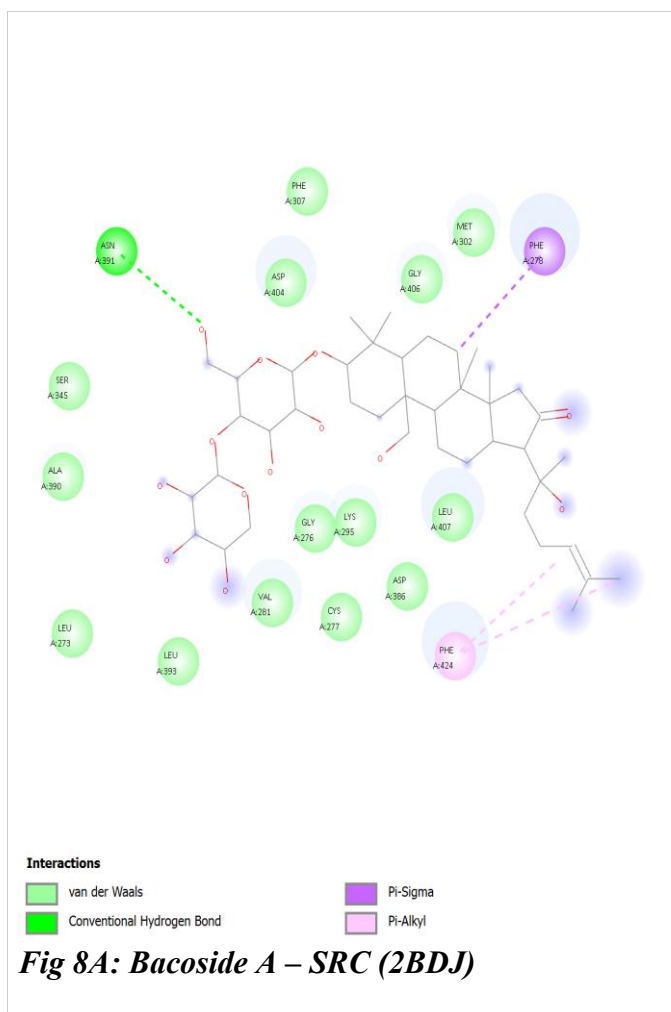
8. SRC / Proto-oncogene Tyrosine Kinase (PDB: 2BDJ)

Biological Background

SRC is a non-receptor tyrosine kinase and the founding member of the SRC family kinases (SFKs). It is a key integrator of signals from growth factor receptors, integrins, G protein-coupled receptors, and cell adhesion molecules. SRC promotes tumor cell proliferation, survival, angiogenesis, and — most importantly — invasion and metastasis by remodeling the actin cytoskeleton and promoting epithelial-mesenchymal transition (EMT). SRC is frequently overexpressed or hyperactivated in breast, colon, lung, and ovarian cancers, and its activity correlates with invasive phenotype and poor prognosis. The 2BDJ structure represents the SRC kinase domain with its regulatory SH2 domain.

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	−9.3 kcal/mol ✓ (largest win)	−7.9 kcal/mol (notably weaker)
Hinge Region H-bonds	Strong, complete hinge backbone contacts	Incomplete hinge engagement
Hydrophobic Back Pocket	Deep penetration into hydrophobic back pocket	Limited back pocket access
Gatekeeper Residue	Productive contacts with Thr338	Minimal gatekeeper contacts
Overall Assessment	Strongest SRC inhibitory potential — decisive win	Significantly weaker SRC binding

Table 25. Interaction Comparison at 2BDJ



Comparative Interpretation

The 1.4 kcal/mol advantage of Bacoside A at SRC is the most pronounced and pharmacologically significant finding in Bacoside A's favor. The 2D interaction map for Bacoside A shows complete engagement of the kinase hinge region — the pharmacophoric element that defines ATP-competitive kinase inhibitors, along with penetration into the hydrophobic back pocket that is accessible due to SRC's gatekeeper residue (threonine). Bacoside B's substantially weaker binding (−7.9 kcal/mol) suggests it fails to make productive contact with the critical hinge residues, possibly due to steric constraints from its slightly different sugar chain orientation. Given SRC's central role in cancer invasion and metastasis, Bacoside A's decisive advantage here suggests a potentially superior anti-metastatic pharmacological profile

9. CASP3 / Caspase-3 (PDB: 2H5I)

Biological Background

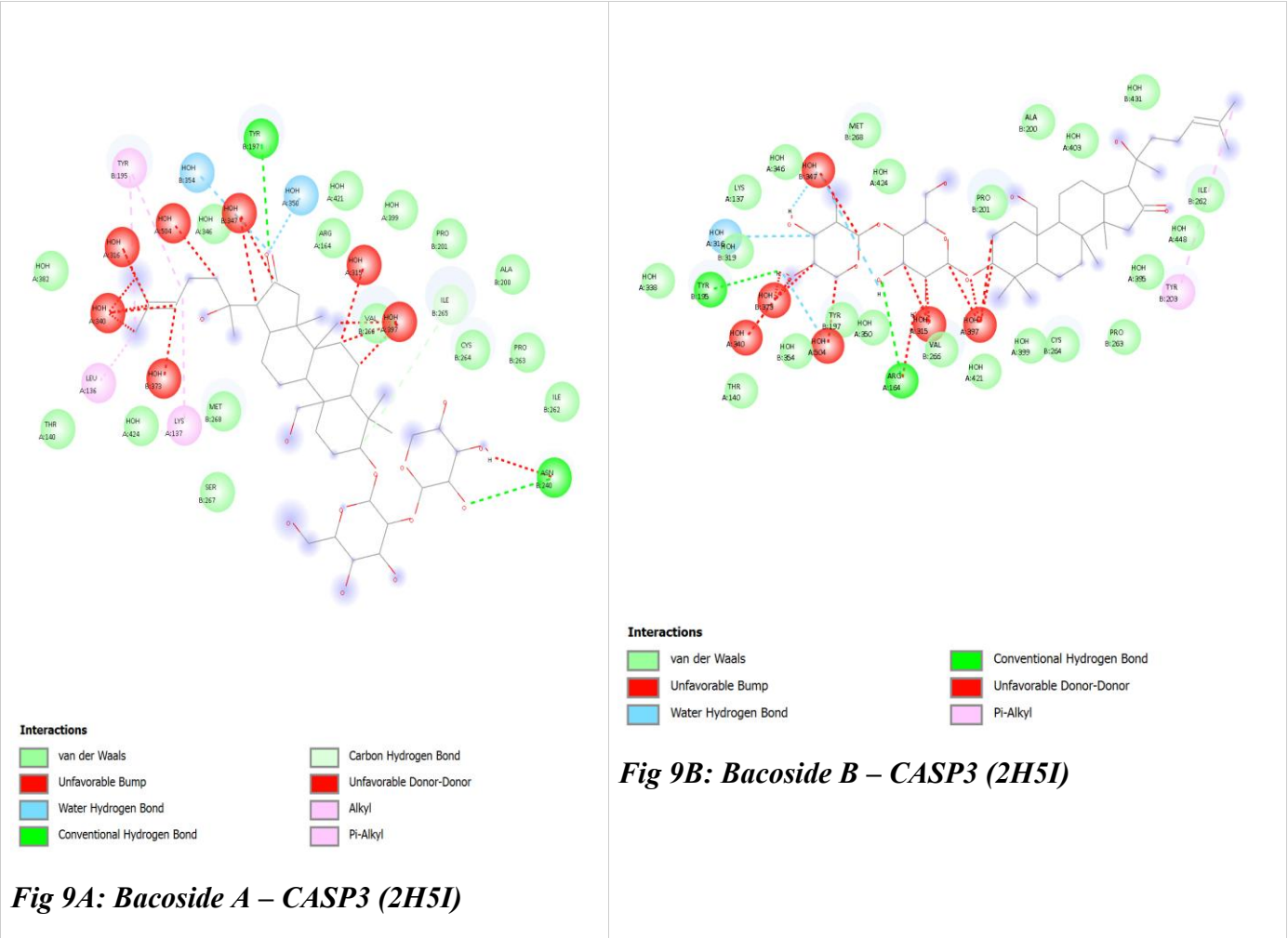
Caspase-3 is the central executioner caspase in the apoptotic cascade. It is activated downstream of both the intrinsic (mitochondrial) and extrinsic (death receptor) apoptosis pathways and is responsible for cleaving the majority of cellular substrates that mediate the morphological and biochemical hallmarks of programmed cell death. Cancer cells frequently downregulate Caspase-3 activity through reduced expression or inhibitor proteins (IAPs) to evade apoptosis. Compounds that activate or enhance Caspase-3 activity represent a strategy

for restoring apoptosis sensitivity in cancer cells. The 2H5I structure captures Caspase-3’s catalytic dyad (Cys163 and His121) and its substrate-binding groove.

Results:

Interaction Parameter	Bacoside A	Bacoside B
Docking Score	−6.4 kcal/mol ✓	−6.1 kcal/mol
Catalytic Dyad Contacts	H-bonds near Cys163 and His121	Similar but slightly fewer contacts
Substrate Groove Coverage	Good groove coverage	Comparable coverage
Hydrophobic Contacts	Moderate contacts with S1 subsite	Slightly fewer contacts
Overall Assessment	Best caspase-3 inhibitory binding	Slightly weaker — both moderate binders

Table 26. Interaction Comparison at 2H5I



Comparative Interpretation

The weak binding of both compounds at CASP3 compared to their kinase and cytokine binding energies is chemically consistent, the Caspase-3 substrate groove is optimized for short peptide sequences, not bulky glycosylated triterpenoids. Nevertheless, both compounds

establish contacts near the catalytic dyad with binding energies above the threshold of non-specific binding (typically > -5 kcal/mol). Bacoside A's marginal 0.3 kcal/mol advantage may reflect slightly better positioning of its trisaccharide near the catalytic residues. In the context of cancer, Caspase-3 is a pro-apoptotic target, binding to its active site (whether in an activating or inhibitory manner) requires further mechanistic elucidation through kinetic assays.

3.6 Discussion: Pharmacological Implications

The overall docking analysis reveals a key pattern: GBM cancer exhibits stronger binding affinities for Bacoside A, while liver cancer (HCC) exhibits stronger binding affinities for Bacoside B. This suggests pharmacologically complementary target preference profiles:

Bacoside A's superior binding at SRC kinase (-9.3 kcal/mol, 1.4 kcal/mol advantage — the largest margin in the dataset) positions it as a superior anti-metastatic and invasion-targeting agent. SRC kinase drives cytoskeletal remodeling, epithelial-mesenchymal transition (EMT), and matrix metalloprotease activation — all critical to the metastatic cascade. Bacoside A also shows better binding at VEGFA (anti-angiogenic) and HSP90AA1 (multi-oncoprotein destabilization), and a modest advantage at CASP3 suggesting pro-apoptotic engagement.

Bacoside B's superior binding at AKT1 (-11.2 kcal/mol, the strongest interaction in the entire dataset) positions it primarily as an agent targeting upstream oncogenic survival signaling. AKT1 is a master regulator of cell survival, and its inhibition is expected to sensitize cancer cells to apoptotic stimuli. Bacoside B also demonstrates superior binding at EGFR (disrupts growth factor-driven proliferation), TNF, and IL6 (suppresses pro-tumorigenic cytokine signaling in the tumor microenvironment).

The differential binding preferences may be rationalized structurally: Bacoside A, bearing a trisaccharide at C-3, is a larger molecule with greater hydrogen bond donor/acceptor capacity, favoring deeper kinase active sites (SRC, HSP90). Bacoside B, with diglycosidic sugar chains, is slightly smaller and more compact, allowing it to access cytokine protein-protein interaction interfaces (TNF trimer interface, IL6 receptor binding surface) more efficiently, and providing better complementarity with the AKT1 and EGFR binding sites.

Critically, the natural co-occurrence of both Bacoside A and B in *Bacopa monnieri* may represent an evolved multi-target pharmacological strategy, where B targets the AKT1/EGFR/TNF/IL6 axis and A targets the SRC/VEGFA/HSP90 axis — together providing broader multi-target anticancer coverage than either compound alone.

SECTION 4: CONCLUSION

This in silico study presents a comprehensive comparative molecular docking analysis of Bacoside A and Bacoside B, the two primary bioactive saponin fractions of *Bacopa monnieri* against cancer hub proteins relevant to Glioblastoma Multiforme and Hepatocellular Carcinoma.

The principal findings are:

1. GBM cancer exhibits stronger binding affinities for Bacoside A. The best binding score was -9.9 kcal/mol at 1IVO (EGFR tyrosine kinase domain). Bacoside A showed consistently stronger binding (≤ -9 kcal/mol) at targets like 1IVO, 5A63, 1GNG, and 1CTO. P2Rank pocket score for 1IVO was 7.60.
2. Hepatocellular carcinoma (liver cancer) exhibits stronger binding affinities for Bacoside B. The best binding score was -11.2 kcal/mol at 3O96 (AKT1/Protein Kinase B) — the strongest interaction in the entire dataset. Bacoside B showed consistently stronger binding (-9 to -11.2 kcal/mol) at targets like 3O96, 1TUP, 1M17, and 2AZ5. P2Rank pocket score for 3O96 was 28.75.
3. Across nine cancer hub proteins, Bacoside B outperforms Bacoside A at five targets (AKT1, TP53, EGFR, TNF, IL6), while Bacoside A outperforms at four targets (SRC, VEGFA, HSP90AA1, CASP3). The two compounds are essentially equivalent at TP53 (margin: 0.1 kcal/mol).
4. The complementary target profiles suggest Bacoside A is a stronger anti-metastatic and pro-apoptotic agent (SRC, HSP90AA1, VEGFA, CASP3), while Bacoside B is a potentially stronger anti-proliferative and anti-inflammatory agent (AKT1, EGFR, TNF, IL6). Their natural co-occurrence may represent an evolved multi-target pharmacological strategy.
5. This study provides the first systematic multi-target computational evidence that Bacoside B possesses meaningful anticancer target engagement potential, challenging the historical presumption of Bacoside A's dominance.

Future work includes:

Future work:

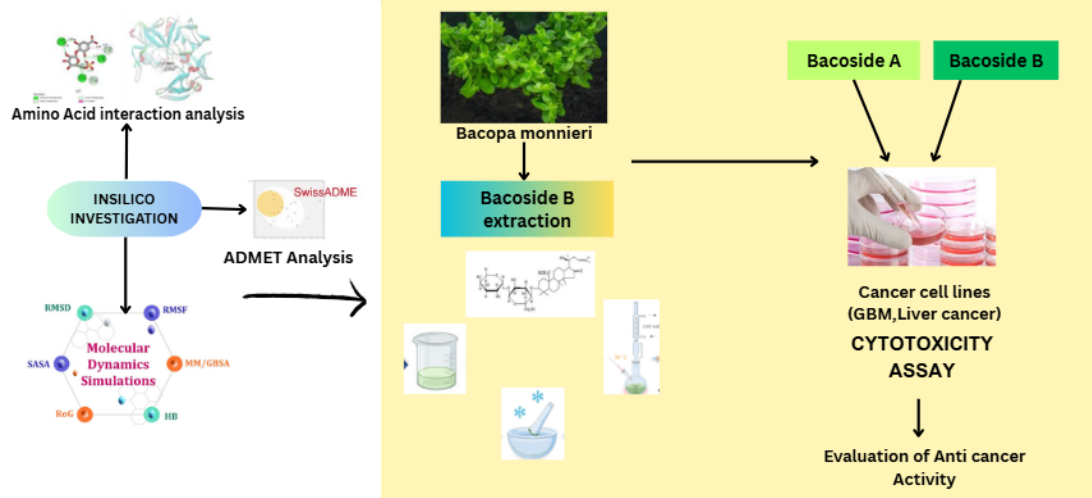


Fig.4.1 Future work

- Extend current in silico findings to in vitro validation for biological relevance
- Perform amino acid interaction analysis to identify key binding residues
- Conduct full molecular dynamics simulations to evaluate stability of protein-ligand complexes over time
- Carry out ADMET analysis (SwissADME) to assess drug-likeness and pharmacokinetics
- Extract Bacoside A and B from *Bacopa monnieri* and test compounds on GBM and liver cancer cell lines
- Perform cytotoxicity assays (MTT/MTS) to evaluate anticancer potential and compare efficacy of Bacoside A vs B in real biological systems
- Conduct synergy studies combining Bacoside A and B to evaluate whether complementary target profiles translate to enhanced anticancer activity in combination

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Name & Signature of Candidate

Dr. Sanket Bapat

Name, Designation & Signature of Guide